Projection of Adoption Trajectories for Diesel and Alternative Powertrain Technologies for Heavy-Duty Class 8 Vehicles in a Line-Haul Regional Network

Ana Guerrero de la Peña[a], Navindran Davendralingam[b], Ali K. Raz[b], Daniel DeLaurentis[b], Gregory Shaver[a], Vivek Sujan[c], Neera Jain[a,*]

[a]*School of Mechanical Engineering, Purdue University, West Lafayette, IN, 47097 USA*
[b]*School of Aeronautics and Astronautics, Purdue University, West Lafayette,IN, 47097 USA*
[c]*Cummins, Inc., Columbus, IN, 47201 USA*

Abstract

In this paper we present a mixed-integer linear program to represent the decision-making process for heterogeneous fleets selecting vehicles and allocating them on freight delivery routes to minimize total cost of ownership. This formulation is implemented to project alternative powertrain technology adoption and utilization trends for a set of line-haul fleets operating on a regional network. Alternative powertrain technologies include compressed and liquefied natural gas engines, diesel-electric hybrid, battery electric, and hydrogen fuel cell. Future policies, economic factors, and availability of fueling and charging infrastructure are input assumptions to the outlined framework. Powertrain technology adoption, vehicle utilization, and resulting CO_2 emissions predictions for a hypothetical, representative regional highway network are illustrated. A design of experiments is used to quantify sensitivity of adoption outcomes to variation in vehicle performance parameters, fuel costs, economic incentives, and fueling and charging infrastructure considerations. Three mixed-adoption scenarios, including BE, HFC, and CNG vehicle market penetration, are identified by the DOE study that demonstrate the potential to reduce cumulative CO_2 emissions by more than 25% throughout the period of study.

1. Introduction

In order to reduce greenhouse gas emissions produced during freight transportation, the way goods are moved, and the technologies used to do so, must be improved. Trucks are the dominant mode of transportation for freight in the U.S. They transport cargo from ports to inter-modal centers, warehouses and retail stores across cities and highways, and to our homes. Trucks move more than 13 billion tons of freight annually, representing more than 80% of all freight movement by weight in the U.S. (U.S. Department of Transportation BTS, 2016; U.S. Department of Transportation, 2017; Hillestad et al., 2009). Since 1990, greenhouse gas emissions from freight trucking have increased five times faster than emissions produced from passenger travel. These numbers are expected to increase by more than 40% by 2040 (U.S. Department of Transportation, 2017; Hillestad et al., 2009).

Heavy-duty Class 7 and 8 trucks, those weighing over 26,000 pounds, represent more than 70% of the vehicle miles traveled by all freight transportation trucks in the U.S. (Hillestad et al., 2009). In recent decades, introduction of the National Highway Traffic Safety Administration's Corporate Average Fuel Economy (CAFE) standards, the EPA greenhouse gas (GHG) emissions regulations, and incentive programs including SmartWay certification, have resulted in a reduction in both fuel consumption and CO_2 emissions (International Council on Clean Transportation, 2017; Transportation Research Board and National Cooperative Freight Research Program, 2010, 2011). This has been achieved by requiring manufacturers to increase average fuel economy of vehicles sold in the U.S. However, an overwhelming majority of the heavy-duty Class 8 vehicles in the country use diesel engines, with less than 2% using natural gas or other alternative fuels (Reinhart, 2016). In 2012, heavy-duty trucks alone emitted more than 70% of the CO_2 emissions in the U.S. freight transportation system (U.S. Department of Transportation BTS, 2016; U.S. Department of Transportation,

*Corresponding author
Email addresses: guerrer2@purdue.edu (Ana Guerrero de la Peña), davendra@purdue.edu (Navindran Davendralingam), akraz@purdue.edu (Ali K. Raz), ddelaure@purdue.edu (Daniel DeLaurentis), gshaver@purdue.edu (Gregory Shaver), vivek.a.sujan@cummins.com (Vivek Sujan), neerajain@purdue.edu (Neera Jain)

2017) as approximately 22.4 pounds of CO_2 are produced per gallon of diesel fuel combusted (U.S. Energy Information Administration, 2017). Alternative powertrain technologies, including compressed (CNG) and liquefied natural gas (LNG) engines, hybridization, battery electric, and hydrogen fuel cells (HFC), have been proposed for energy savings and emission reductions. These technologies, however, may require revolutionary changes to vehicle operations and infrastructure. For example, a wide network of hydrogen fueling and electric charging stations does not presently exist, and vehicle range and payload capacity may decrease given adoption of alternative powertrain options when compared to diesel. Moreover, there is uncertainty as to how these emerging technologies will impact vehicle reliability and what policies will be introduced to regulate or incentivize their adoption.

Economic value propositions must be evaluated in terms of initial purchasing or up-charge costs, as well as reduced operational costs, in order to determine their attractiveness and potential for market adoption. A large majority, 86%, of Class 8 tractors in the U.S. are company-owned and operated as a *fleet*, whereas only 14% are owner-operator trucks (Transportation Research Board and National Cooperative Freight Research Program, 2010). As a result, the commercial vehicle market is driven by knowledgeable customers who focus on the efficiency of their operations, including the fuel costs associated with accomplishing their tasks. A tool that can be used to understand effects of freight system evolution on fleet adoption behaviors and market penetration of new technologies can help manufacturers, and policy makers alike, design and introduce products and policies to maximize adoption of cleaner technologies that target desired emissions outcomes.

The search for a strategic transition to cleaner trucks is not new in the literature. However, few studies have focused on modeling the way fleets, particularly Class 8 fleets, adopt vehicles based on their perceived economic attractiveness *as the freight transportation system (FTS) evolves over time*. Table 1 summarizes the literature on emerging technology adoption, indicating that current studies only account for variables within a few domains (e.g. vehicle performance, single fleet decisions, etc.), while assuming all other FTS considerations, e.g. number of vehicles, vehicle miles traveled and other utilization metrics, energy costs, etc., remain indefinitely constant. These assumptions, while at times necessary to manage scalability of the problem, mask the interacting effects of the FTS evolution on fleet adoption behaviors and can lead to an incomplete study of market adoption viability.

Table 1: FTS Gaps in Literature Summary

Study	Vehicle cost analysis	Fleet utility analysis	Policy effects on adoption & emissions	Heterogeneous market composition & emissions	Effects of system evolution	Vehicle allocation & utilization trends
Davis and Figliozzi (2013)	X	X	X			
Delorme et al. (2010)	X					
Feng and Figliozzi (2012)	X	X	X			
Fulton and Miller (2015)	X			X		
Graham et al. (2008)				X		
Janic (2007)		X				
Kleiner et al. (2015)	X	X				
Lammert et al. (2014)	X					
Moultak et al. (2017)	X	X		X		
Schäfer and Jacoby (2006)	X			X		
TRB and NCFRP (2010)	X					
Wadud et al. (2016)	X		X			
Zhao et al. (2013b)	X	X				

The majority of published studies focus on projecting the competitiveness of emerging vehicle technologies in terms of fuel savings by evaluating performance and emissions over *pre-determined drive-cycles*. However, this approach does not account for other FTS considerations, including infrastructure evolution, policy-related incentives and operational restrictions, and their effects on vehicle operation and fleet behaviors. For example, Delorme et al. (2010) present a simulation-based assessment to show a 3-17% fuel savings for mild and full-hybrid electric diesel Class 8 vehicles with battery capacity ranging from 5-25 kWh given specified hybrid configurations and route types. Zhao et al. (2013b) present a fuel savings comparison for diesel, LNG, hybrid electric, battery electric, and fuel cell heavy-duty vehicle architectures for selected day, short-haul, and long-haul drive cycles as a measure of the economic attractiveness of alternative technologies. Their study demonstrates a CO_2 emissions reduction between 24-39% for day drive cycles and 12-29% reduction for short and long-haul cycles for vehicles with alternative powertrain configurations. Lammert et al. (2014) present fuel savings for two diesel-powered platooning Class 8 tractor-trailer vehicles operating on simplified drive-cycles given varying gap distances, steady-state speeds, and gross vehicle weight to demonstrate technology attractiveness.

Several studies present single-fleet vehicle ownership cost minimization models which consider cost of acquisition and operational costs for fixed routes. Davis and Figliozzi (2013) introduce a series of scenarios for which adoption of electric vehicles offer a reduction in cost of ownership, when compared to diesel adoption, for a single mixed-composition delivery fleet over an 11-year period. Feng and Figliozzi (2012) present a similar diesel and electric delivery vehicle adoption model for a single mixed-fleet, additionally introducing replacement cycles and vehicle salvage revenue. While both studies evaluate ownership cost sensitivity to vehicle speed, battery life, and tax incentives, they do so by imposing fixed route lengths and daily vehicle utilization. The aforementioned studies do not consider the availability of charging infrastructure and route optimization, thereby ignoring the relationship between infrastructure system evolution and vehicle operation planning.

Other studies project CO_2 emissions resulting from assumed market penetration scenarios and extrapolation of historical data, including projected number of vehicles and annual vehicle miles traveled (VMT). Fulton and Miller (2015) demonstrate pathways to achieve 80% reduction of CO_2 emissions in the U.S. by 2050 by assuming high market penetration scenarios for low-carbon fuel architectures and electric vehicles. Similarly, Schäfer and Jacoby (2006) present adoption rate scenarios for personal vehicles and heavy-duty Class 8 diesel trucks while considering economic penalties imposed on CO_2 emissions. However, while these studies describe projected emissions levels for given scenarios, they do not provide stakeholders with any insight on the *FTS mechanisms necessary to achieve* the vehicle adoption rates proposed.

The future adoption rates and mixed technology composition of the line-haul FTS will be a result of the individual operational strategies of the many fleets that service it, given that they are collectively influenced by the evolution of their common environment. Individual fleets of varying characteristics will maintain their independence by self-regulating their adoption cycles and vehicle utilization strategies all the while contributing and taking resources from other FTS systems (e.g., road network, traffic, freight demand, policy incentives) (Maier, 1998). System-of-Systems Engineering (SoSE) (Jamshidi, 2008) is a specialized field of research that addresses the emerging capabilities resulting from interactions between the independent and distributed systems that constitute large-scale complex systems. SoSE concepts and methodologies offer significant value to the study of technology adoption rates and emissions at a large scale, i.e. a line-haul FTS, resulting from the behaviors of the multiple independent and distributed fleets operating in such a system. Researchers are beginning to recognize the SoS framework to be the foundational element for the study of transportation related problems, including traffic control (Aliubavicius et al., 2016), real-time scheduling of dangerous goods transportation (Roncoli et al., 2013), and innovations for future intelligent transportation systems (Mostafavi et al., 2011).

In recent work (Guerrero et al., 2018), the authors developed a model of fleet vehicle purchasing behaviors by modeling the FTS as an SoS (DeLaurentis, 2005) and defining a representative set of heterogeneous fleets to project historical technology adoption trends in the line-haul segment. Here we extend that framework to capture the evolution of line-haul freight transportation system factors and their influence on the future economic attractiveness of *emerging and alternative* vehicle powertrain technologies for freight transportation. In particular, the framework here proposed contributes a more mathematically complex model considering the operational challenges of emerging powertrains including natural gas, battery electric, and hydrogen fuel cells. This is accomplished by including the effects of vehicle range and performance, fleet-level considerations, infrastructure availability, and policies on fleet-wide operational and purchasing decisions. The proposed model formulation is implemented to project vehicle powertrain technology adoption and allocation for multiple heterogeneous fleets operating in a regional freight transportation network and the resulting CO_2 emissions trajectories in the region of interest. The proposed model is capable of evaluating the sensitivity of projected adoption trends and resulting CO_2 emissions to variation of vehicle, policy, and infrastructure design parameters. Furthermore, it provides an understanding of the technology design and performance, policies and incentives, and availability of infrastructure needed to achieve the adoption and emission effects desired in the future. The systematic approach followed in this manuscript to define, abstract, and implement a model of the FTS is presented in Figure 1.

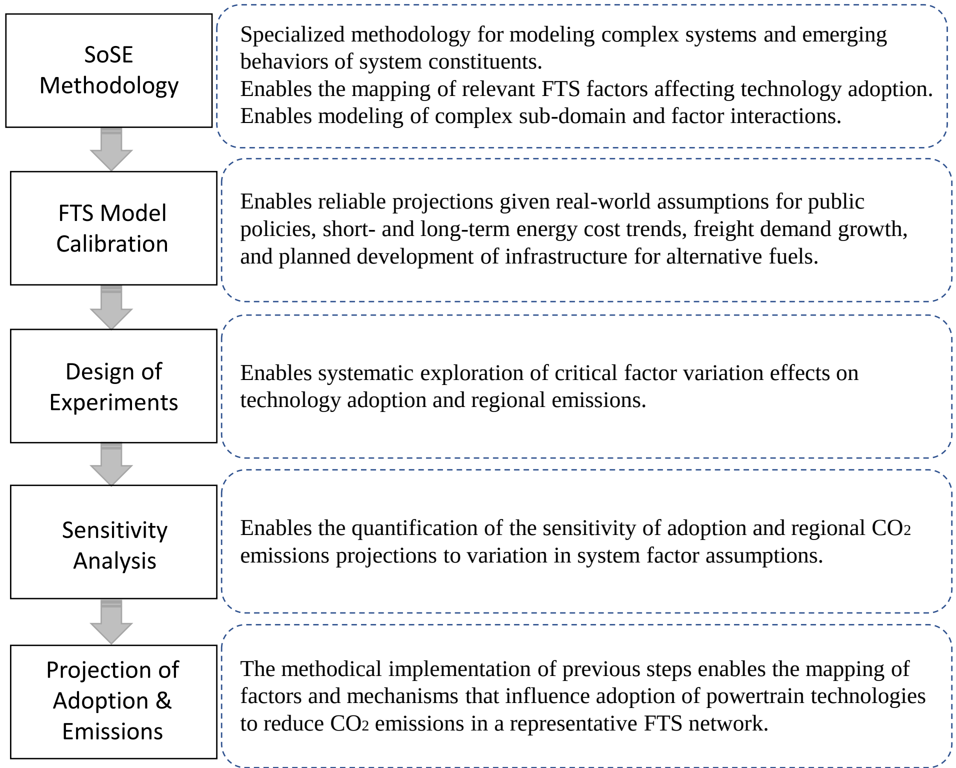


Figure 1: Methodology presented to define, abstract, and implement a model of a regional FTS to project adoption of powertrain technologies for line-haul vehicles.

This paper is organized as follows. Section II presents an overview of the proposed model scope and abstraction for a regional line-haul freight transportation system-of-systems. Section III describes the linear program formulation proposed to model mixed-fleet vehicle adoption and allocation behaviors based on operational and vehicle purchasing costs. Section IV presents a sensitivity analysis of the framework to different system factors, including vehicle performance parameters, fuel costs, infrastructure availability, and policies, among others. Concluding remarks are made in Section V.

2. Freight Transportation as a System of Systems

To model the U.S. line-haul freight transportation system as a system-of-systems, we employ a methodology developed by DeLaurentis (2005) consisting of three phases: definition, abstraction, and implementation. The purpose of the *definition phase* is to decompose the SoS problem across two dimensions, therefore mapping the scope of relevant system factors and their critical interactions on a Resources, Operations, Policies, and Economics (ROPE) matrix (DeLaurentis and Callaway, 2004) across the relevant hierarchies of the FTS. Assisted by the ROPE matrix, the *abstraction phase* identifies the sub-domains of the FTS and the flow of information between them. Once these have been defined, the appropriate level of abstraction necessary to model variables and sub-domains can be identified. Finally, the *implementation phase* realizes the abstracted SoS framework within a modeling and simulation environment. Additional details of this particular SoS methodology are available in (DeLaurentis, 2005). Here we discuss its application to the present focus of modeling emerging powertrain technology adoption in the U.S. line-haul FTS.

2.1. Definition

The ROPE matrix defined to represent the U.S. line-haul FTS is shown in Table 2. Here, focus is placed on those factors and considerations of the FTS that affect the adoption trends of powertrain technologies. The first dimension identifies the hierarchical levels of the SoS. The different powertrain options considered—diesel, CNG, LNG, hybrid electric diesel, battery electric, and hydrogen fuel cell—are introduced as the α level resources of the SoS. The engine, chassis, and body of the trucks are often produced by different manufacturers but are integrated as a single vehicle. As such, fuel consumption and economic attractiveness must be determined by the features of the complete vehicle architecture (Transportation Research Board and National Cooperative Freight Research Program, 2010; Zhao et al., 2013b). For this reason, the next level of the hierarchy, the β-level, represents the heavy-duty Class 8 vehicle architectures encompassing the powertrain technologies introduced. The γ-level includes the set of vehicles that comprise a single regional mixed-adoption fleet. Finally, the δ-level is defined as the regional freight transportation network—the nodes, infrastructure, and highway routes over which a collection of line-haul fleets will operate to satisfy freight demand in the region.

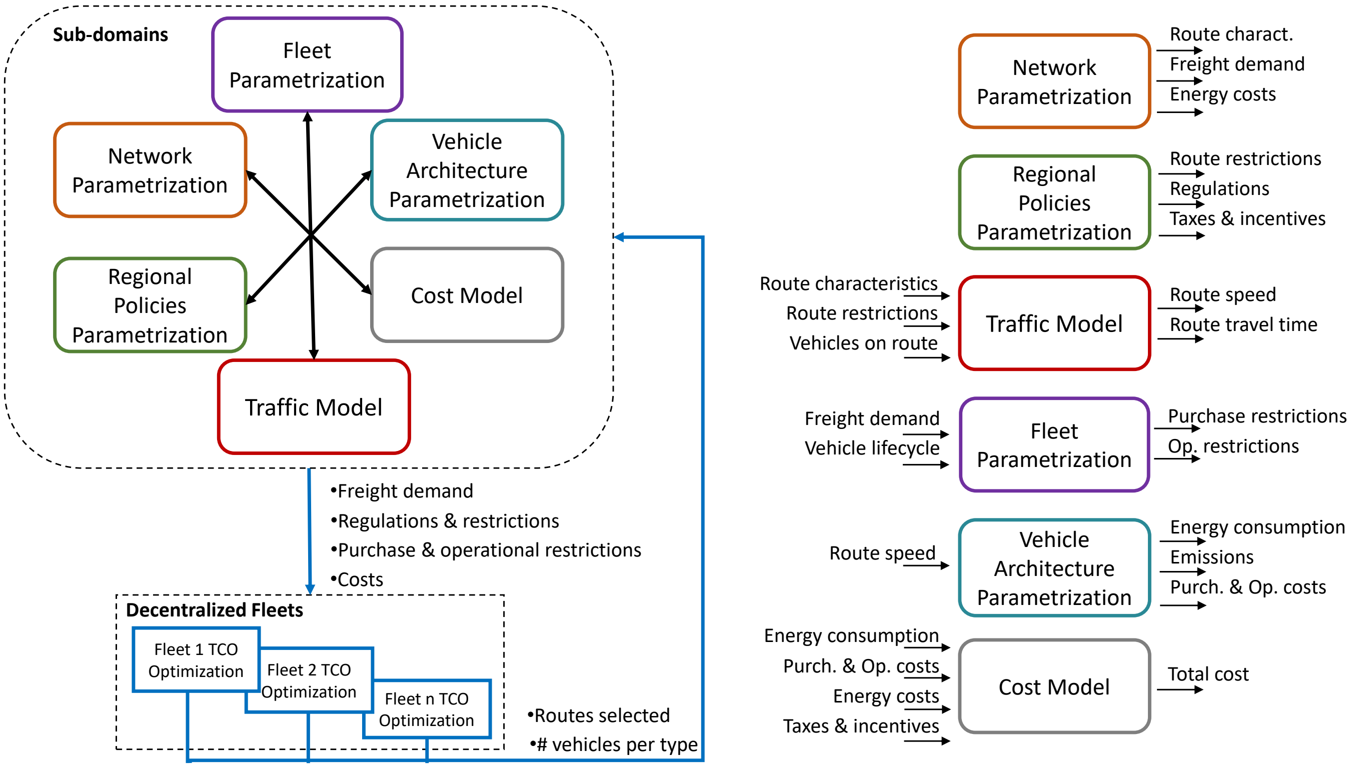
The second dimension of the ROPE matrix spans the breadth of the SoS space across every hierarchical level, defining the relevant resource, operational, economic, and policy factors that must be considered. For example, at the α and β levels, policies regarding powertrain greenhouse gas emissions and vehicle weight limits will influence operational factors including vehicle fuel consumption and cargo capacity. At the γ level, relevant economic considerations including total cost of ownership (TCO), tax and cash incentives, and cost of use of infrastructure influence a fleet's decision regarding the vehicle architectures they select. Finally, it is by inclusion of operational, economic and policy factors at the δ level that we can observe the collective impact of individual and isolated decisions in the SoS. The particular set of policies that are active in the region, whether to regulate highway speed limits or to provide economic incentives for cleaner technologies, among others, will influence the attractiveness of the technologies introduced and fleet adoption behaviors in a particular manner. For example, adoption trends for different vehicle technologies cannot be observed only at the powertrain, α, and vehicle, β, levels as they are dominated by the operations, economics, and policies at the fleet and regional network levels of the hierarchy.

Table 2: Regional Line-Haul Freight Transportation System ROPE matrix

Level	Resources	Operations	Policy	Economics
Alpha	-Powertrain Type: Diesel Natural gas (CNG,LNG) Hybrid electric Battery electric Hydrogen fuel cell	-Powertrain fuel consumption -Powertrain reliability -Maintenance	-Emission restrictions -GHG regulations	-Cost of fuel -Cost of energy -Cost of maintenance
Beta	-Heavy duty Class 8 vehicles -Vehicle architecture	-Ton-mi/gal efficiency -Average day operation -Vehicle life cycle -Cargo load/capacity -Miles driven based on selected routes -Operate at constant speed over route -Vehicle range -Fueling/charging time	-80,000 lb weight limit -Tax or purchase incentives	-Cost of fuel/energy consumed -Cost of driver/hour -Cost of maintenance -Cost of purchase -Cost of equipment
Gamma	-Vehicles in single regional fleet	-Freight distribution -Fleet size -Vehicle replacement cycles and years of service limits -Operator hours -Fuel/charging stops -Return-to-base operation	-Driver 11 hours of service limits -14 hr window of operation	-Total cost of ownership decision metrics per year -Tax and cash incentives -Cost to use/own infrastructure
Delta	-Regional highway network: Cities 1-4 Distribution centers 1, 2 Highways, roads and supporting infrastructure	-Total freight demand between cities by weight -Traffic conditions: vehicles on road, road density capacity, travel time -Availability of fueling/charging stations -Availability of on-road charging	-Route speed limit -Regional emissions restrictions -Regional incentive programs	-Cost of fuel/electricity -Charges for use of infrastructure

2.2. *Abstraction*

Based on the above problem scope and the SoS definition of the line-haul FTS, the abstraction phase identifies the interconnections and sub-domains of this problem space such that models and their input and output parameters and interactions can be abstracted, as shown in Figure 2.



(a) Abstraction into sub-domains and decentralized fleets.

(b) Flow of information between sub-domains.

Figure 2: Abstraction of the regional line-haul freight transportation system.

The abstraction model shown in Figure 2a represents the influence of system sub-domains, in the form of rules, restrictions, and constraints, on the decision process of multiple *decentralized* fleets to purchase those vehicle architectures that are economically attractive to them given their operational and purchase costs. On this diagram we observe that a single fleet acts with *centralized control of their vehicles*; that is to say, the fleet, acting as a single stakeholder, will not only select the vehicles they adopt but also the routes over which their vehicles are operated. Furthermore, multiple fleets will operate independently of each other over a single region, each with similar centralized control over their vehicles. Technology composition may vary across the many fleets that service the FTS given their individual management and operational strategies, including vehicle replacement cycles and annual fleet growth, as they are collectively influenced by their common environment. Allocation of the several independent fleets over the same network will have an effect on traffic, which in turn may affect route selection and operating costs for all fleets. As will be shown in Section 3, this fleet purchasing and vehicle deployment behavior can be represented by an optimization problem that can be solved via mixed-integer linear programming (MILP).

2.3. Implementation

In order to predict the future adoption of powertrain technologies across a set of representative FTS line-haul fleets, the sub-domains described earlier are first calibrated to represent a region of interest and then solve the MILP as shown in Figure 3. Fleet parameters are calibrated to represent 12 fleets with varying cargo transportation targets, annual growth, and management characteristics. Vehicle technology adoption and utilization trends are optimized (via the MILP) annually across each individual fleet in order to minimize their TCO. The total number of vehicles allocated by each fleet on the FTS network routes is then used to compute the traffic flow speed conditions for the following year. This process continues over the duration of the simulation period. The projection of both powertrain technology adoption and vehicle use also enables the computation of expected CO_2 emissions. The proposed model is implemented in MATLAB 2016b, as shown in Figure 3, with the use of the YALMIP toolbox (Löfberg, 2012) and the Gurobi Optimizer 7.5.1 MILP solver (Gurobi Optimization, Inc., 2016) as described by Guerrero et al. (2018).

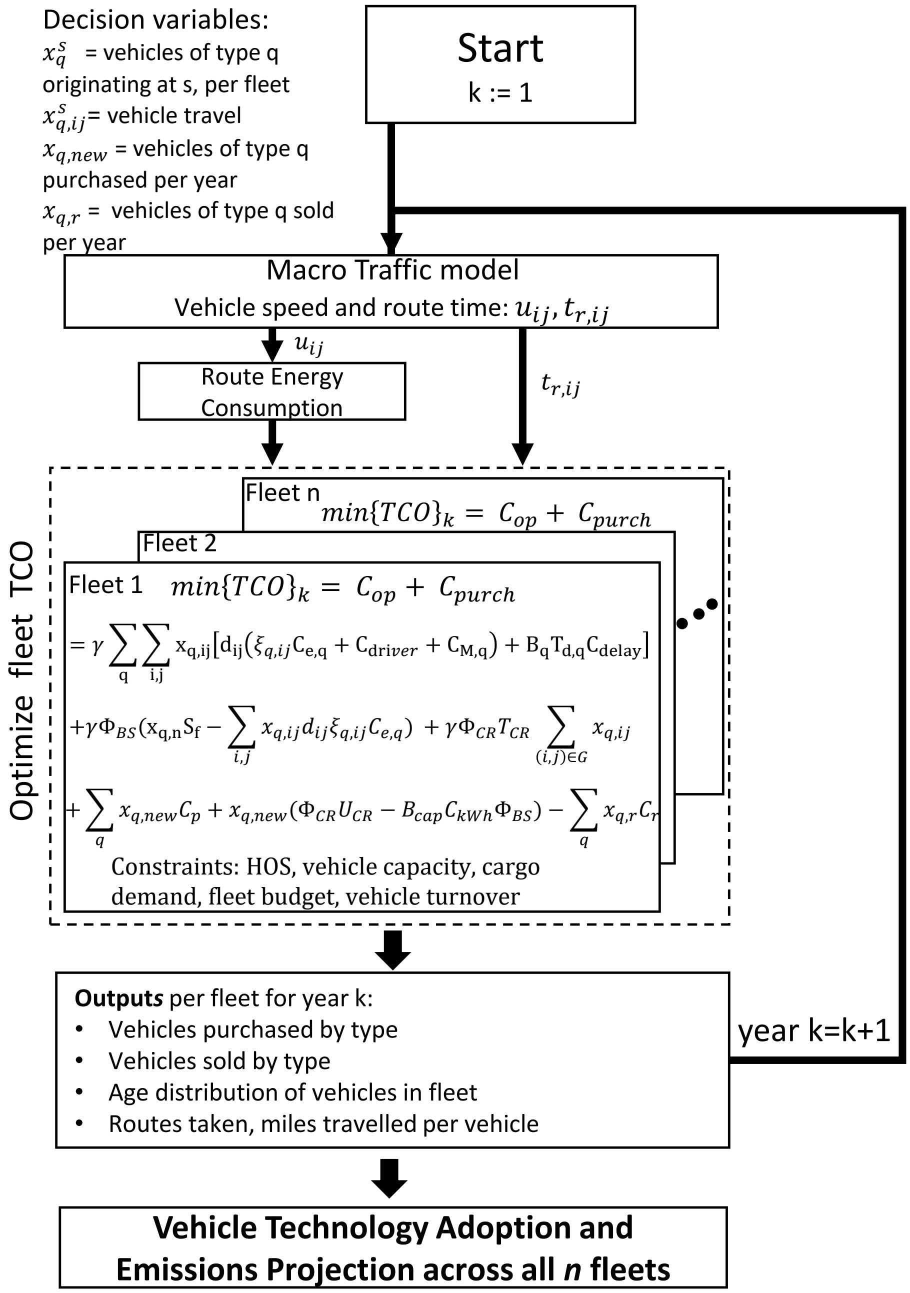


Figure 3: Implementation of the mixed-integer linear program to project vehicle technology adoption across regional fleets.

3. Fleet TCO Optimization

Here we describe the mixed-integer linear program, the objective function and constraints, used to predict a single fleet's technology adoption and utilization behaviors. A system (fleet-wide) optimal traffic allocation approach is used to project vehicle utilization and compute operational costs, while the annual fleet acquisition costs are comprised of new vehicle purchases and offset by vehicle turnover sales revenue. Fleet management, policies, and vehicle operational considerations with respect to energy consumption, freight demand, vehicle capacity, and infrastructure availability, among others, are formulated as constraints. This optimization program is implemented in parallel for multiple fleets operating over a single regional network as shown in Figure 3. Tables 3-5 describe the sets, decision variables, and parameters used to formulate the optimization program.

Table 3: Set Definition

Sets	Description
A	Set of traversable links between city nodes
G	Set of route links with wireless charging capability
H	Set of cargo origins
N	Vehicle enumeration
Q	Set of vehicle architecture types
S	Set of vehicle origins

Table 4: Variable Definition

Decision Variables	Description
x^s_{qn}	Number n of vehicles type q that originate at s per year
$x^s_{qn,ij}$	Vehicle n of type q originating at *s*, traveling on link (i,j).
y^h_{ij}	Cargo flow between links, originating at *h*.
$x_{q,new}$	Number of new vehicles of type *q* purchased per year.
$x_{q,r}$	Number of vehicles of type *q* sold per year.
$e_{q,ij}$	Feasibility of travel for vehicle type *q* over route *(i,j)*.
Φ_{BS}	Binary variable, 1 if battery-electric vehicles use battery swap stations.
Φ_{CR}	Binary variable, 1 if battery-electric vehicles travel on charging roads.
$\Phi_{qn,c}$	Number of times vehicle *n* of type *q* stops to fuel, charge, or perform a battery swap.

Table 5: Parameter Definition

Parameters	Description
γ	Estimated vehicle life-cycle period in years
η	Hours of service window of operation
$\Psi_{ttq,q}$	Tank-to-wheel (tailpipe) CO_2 emissions in kg per energy consumption
$\Psi_{wtt,q}$	Well-to-tank CO_2 emissions in kg per energy consumption
ξ_q	Efficiency function of vehicle type *q* gallons or kWh per mile
B_{cap}	Battery capacity in kWh
B_f	Fleet's annual budget for vehicle purchase
b^h_i	Cargo demand from origin *h* to destination*i* in tons
B_q	Reliability of vehicle architecture *q*, % trips delayed per year
C_{driver}	Cost of driver per mile
$C_{delay,c}$	Revenue loss due to delay for cargo type *c*
$C_{eq,b}$	Cost of electricity used by battery electric vehicles in dollars per kWh
$C_{eq,f}$	Cost of fuel consumed by vehicle type *q* in dollars per gallon
C_{kWh}	Cost of battery in $/kWh
$C_{M,q}$	Cost of maintenance per mile for vehicle architecture *q*
$C_{p,q}$	Cost of purchase for vehicle type *q*
$C_{r,q}$	Resale value for vehicle type *q*
d_{ij}	Length (distance) of link *(i,j)* in miles
E_q	Fuel or Battery charge gained at fueling, charging or battery swap station in kWh
E_{CR}	Battery charge gained when traveling on charging road in kWh
$F_{q,i}$	Availability of infrastructure for vehicle type *q* in city node *i*
h_{os}	Hours of service driving limit
l_{max}	Max number of years until replacement for fleet vehicles
l_{min}	Min number of years until replacement for fleet vehicles
M	A sufficiently large number
R_q	Driving range of vehicle type *q* in miles
S_f	Annual service fee, per vehicle, for use of battery swap service
$t_{r,ij}$	Travel time over route *(i,j)* in hours
T_{CR}	Toll, in $, for use of on-road charging
$T_{d,q}$	Average length of time delay for vehicle architecture *q*
u_{ij}	Vehicle steady-state speed on link *(i,j)* in miles per hour
U_{CR}	Vehicle upcharge cost for wireless road charging capability
W_q	Capacity for vehicle type *q* in tons

3.1. Objective Function

The objective function comprises the most critical cost components used by fleet owners to determine the cost of purchase and cost of operation, and therefore the TCO, over a vehicle's lifecycle. These cost components include cost of acquisition, fuel consumption costs, repair and maintenance, and driver wages (Torrey and Murray, 2016). Each individual fleet will select the type of vehicles they purchase and operate as a strategy to minimize the TCO. This decision-making process is exercised annually throughout the projection period. The objective function is therefore defined as follows: $J = C_{op} + C_{purch}$. The cost of operation includes the cost of energy consumed by all vehicles in the fleet C_{ec}, driver wages C_{wages}, vehicle maintenance C_M, and revenue losses C_R as a function of technology reliability such that $C_{op} = \gamma[C_{ec} + C_{wages} + C_M + C_R]$. In order to estimate lifecycle values, daily operational costs are accrued over the number of years in a fleet's TCO outlook period, γ. Here, γ acts as the planning horizon for which a fleet computes the total operational costs of the vehicles selected. It is fleet specific and may be less than or equal to the projection period.

Energy consumption costs will vary depending on the powertrain technology used, as a result of fuel and electricity costs and vehicle efficiency. In the case of diesel, natural gas, and hydrogen fuel cell vehicles, the use of conventional fueling stations is assumed. The primary differentiator between these fueling stations is the time required to fuel the different powertrains. In the case of battery electric vehicles, (BE), however, three different methods of charging are considered: fast-charging stations, battery swap stations, and on-road charging. The fast-charging stations are analogous to conventional fueling stations. Under the battery swap model, service providers own the vehicle batteries and supply users with access to battery swap infrastructure in exchange for a service fee. This option could possibly reduce vehicle upfront costs and battery replacement costs with a predictable annual fee, while expanding BE vehicle range without jeopardizing time spent at the station (Neubauer and Pesaran, 2013; Zheng et al., 2014).

On the other hand, charging roads, also known as inductive, contactless, or dynamic wireless power transfer systems, can charge battery electric vehicles (BEs) without any physical interconnection. These systems can be installed on roadways in order to charge the vehicles while driving and extend the range or decrease the battery size (Stamati and Bauer, 2013; Suh and Kim, 2013). However, the power transfer capabilities of on-road charging systems are limited and may not be sufficient to fully recharge the large batteries required for line-haul operation. For simplicity, it is assumed that all BE vehicles in a single fleet will use only one mode of charging operation.

The cost of energy for daily operation of a heterogeneous fleet is defined as $C_{ec} = C_{ec,f} + C_{ec,b}$ where

$$C_{ec,f} = \sum_q \sum_{(i,j)\in A} x_{q,ij} d_{ij} \xi_{q,ij} C_{eq} \,, \tag{1a}$$

$$C_{ec,b} = \sum_{(i,j)\in A} x_{q,ij} d_{ij} \xi_{q,ij} C_{eq}(1 - \Phi_{BS}) + x_{qn} S_f \Phi_{BS} + \Phi_{CR} T_{CR} \sum_{(i,j)\in G} x_{q,ij} \,. \tag{1b}$$

Here, $C_{ec,f}$ is the cost of fuel consumed by diesel, natural gas, and hydrogen-powered vehicles. For BE vehicles, a subscription model is assumed if battery swap stations are used ($\Phi_{BS} = 1$), in which case the fleet will pay an annual fee per vehicle, S_f, to the service provider. If on-road charging infrastructure is used ($\Phi_{CR} = 1$), fleets will alternatively incur charges for the electricity consumed and associated tolls, T_f, per trip. No additional fees, other than the cost of energy, are assumed for the use of conventional charging stations. As a result, the cost $C_{ec,b}$ includes the cost of electricity consumed by battery electric vehicles and associated tolls or fees for use of charging infrastructure. The variable $x_{q,ij} = \sum_s \sum_n x^s_{qn,ij}$ represents the flow of vehicles of type q over highway link (i, j) regardless of their origin. The sets A and G represent the set of city-nodes in the network and the set of road links with wireless charging capability, respectively. The efficiency of a vehicle using technology type q, ξ_q, is a function of average vehicle speed over a route (i, j) such that $\xi_{q,ij} = \xi_q(u_{ij})$. Other operational costs are computed as follows:

$$C_{wages} = \sum_q \sum_{(i,j)\in A} x_{q,ij} d_{ij} C_{driver} \,, \tag{2}$$

$$C_M = \sum_q \sum_{(i,j)\in A} x_{q,ij} d_{ij} C_{M,q} \,, \tag{3}$$

$$C_R = \sum_q \sum_{(i,j)\in A} x_{q,ij} B_q T_{d,q} C_{delay,c} \,. \tag{4}$$

The cost of purchase comprises the cost of buying new fleet vehicles and is offset by the revenue generated by vehicle turnover sales: $C_{purch} = C_{nv} - C_{sr}$. Fleets will purchase new vehicles 1) to replace those beyond their optimal economic lifecycle, or 2) to address an increase in freight demand by increasing fleet volumes. It is assumed that all vehicles are purchased new, such that the purchase costs of a mixed-adoption fleet are given by $C_{nv} = C_{nv,f} + C_{nv,b}$ where

$$C_{nv,f} = \sum_q x_{q,new} C_{p,q}\,, \tag{5a}$$

$$C_{nv,b} = x_{q,new}(C_{p,q} - B_{cap} C_{kWh} \Phi_{BS} + \Phi_{CR} U_{CR})\,. \tag{5b}$$

Similarly, as with operational costs, a subscription model is assumed for the use of battery swap stations. Contrary to direct ownership, the service provider owns the battery and thus upfront vehicle costs are reduced given the battery capacity, B_{cap}, and assumed battery cost, C_{kWh}. If on-road charging is selected, electric vehicles must have pick-up winding equipment installed (Stamati and Bauer, 2013), incurring an up-charge cost, U_{CR}. A traditional direct ownership model is assumed for all other vehicle types.

Fleets will sell older vehicles once maintenance and repair costs increase as vehicle performance and efficiency decline (North American Council for Freight Efficiency, 2016; Feng and Figliozzi, 2012). The revenue obtained from vehicle turnover sales is computed as

$$C_{sr} = \sum_q x_{q,r} C_{r,q} \tag{6}$$

and implemented as an offset to the purchasing budget and total fleet TCO for the current year. The variables $x_{q,new}$ and $x_{q,r}$ are introduced to represent the vehicles of technology type q that are adopted and those that are sold for replacement, respectively, in the current year of projection, k. These variables are defined as

$$x_{q,new} = [x_q]_k - [x_q]_{k-1} \tag{7a}$$

$$x_{q,r} = [x_q]_{k-1} - [x_q]_k \tag{7b}$$

where $x_q = \sum_s \sum_n x^s_{qn}$. In summary, the cost function J is defined as the TCO and computed via a combination of Eqs. (1)-(6) as follows:

$$\begin{aligned} J =& \gamma \sum_q \sum_{(i,j)\epsilon A} x_{q,ij}[d_{ij}(\xi_{q,ij} C_{e,q} + C_{driver} + C_{M,q}) \ + B_q T_{d,q} C_{delay,c}] \\ &+ \gamma \Phi_{BS}(x_{q,n} S_f - \sum_{(i,j)\epsilon A} x_{q,ij} d_{ij} \xi_{q,ij} C_{e,q}) + \gamma \Phi_{CR} T_{CR} \sum_{(i,j)\epsilon G} x_{q,ij} \\ &+ \sum_q x_{q,new} C_{p,q} + x_{q,new}(\Phi_{CR} U_{CR} - B_{cap} C_{kWh} \Phi_{BS}) - \sum_q x_{q,r} C_{r,q}\,. \end{aligned} \tag{8}$$

3.2. Constraints

In the optimization problem, constraints are enforced on vehicle purchase, resale, and operation. Equations (9a)-(9e) describe constraints imposed on vehicle allocation, $x^s_{qn,ij}$, based on cargo demand, vehicle capacity, vehicle flow balance, and hours of service regulations as defined in Guerrero et al. (2018).

$$\sum_j y^h_{ji} - \sum_j y^h_{ij} = b^h_i \tag{9a}$$

$$\sum_h y^h_{ij} \le \sum_s \sum_q \sum_n x^s_{qn,ij} W_q \tag{9b}$$

$$\sum_j x^s_{qn,ji} - \sum_j x^s_{qn,ij} \ge 0,\ i \ne s \tag{9c}$$

$$x^s_{qn,ii} = 0,\ \ y^h_{ii} = 0 \tag{9d}$$

$$\sum_{(i,j)\in A} x^s_{qn,ij} t_{r,ij} \le h_{os}, \forall s, q, n \tag{9e}$$

An intermediate binary variable, x^s_{qn}, is introduced such that $x^s_{qn} \leq M \sum_j x^s_{qn,ij}$ and $Mx^s_{qn} \geq \sum_j x_{qn,ij}$ for all $i = s$. This indicates the allocation of the nth vehicle of type q over the network, assisting in the computation of total number of vehicles of type q purchased and originating in city s. In order to determine the number of new vehicles purchased at any given city of origin s during the present year, an intermediate binary variable, u^s_q is introduced as shown in Equations (10a)-(10e), where $x^s_q = \sum_n x^s_{qn}$.

$$[x^s_q]_k - [x^s_q]_{k-1} + M(1 - u^s_q) \geq 0 \tag{10a}$$

$$[x^s_q]_k - [x^s_q]_{k-1} - M(1 - u^s_q) \leq 0 \tag{10b}$$

$$x^s_{q,new} - [x^s_q]_k + [x^s_q]_{k-1} + M(1 - u^s_q) \geq 0 \tag{10c}$$

$$x^s_{q,new} - [x^s_q]_k + [x^s_q]_{k-1} - M(1 - u^s_q) \leq 0 \tag{10d}$$

$$x^s_{q,new} \leq Mu^s_q\,, \quad x^s_{q,new} \geq 0 \tag{10e}$$

New vehicle purchases, $x^s_{q,new}$, and vehicle resale, $x^s_{q,r}$, are constrained as defined by Guerrero et al. (2018), given fleet budget and vehicle age such that

$$\sum_s \sum_q x^s_{q,new} C_{p,q} - \sum_s \sum_q x^s_{q,r} C_{r,q} \leq B_f\,, \tag{11}$$

and

$$x^s_{q,new}(t_{yk} - l_{max}) \leq x^s_{q,r} \leq x^s_{q,new}(t_{yk} - l_{max}) + x^s_{q,new}(t_{yk} - l_{min})\,. \tag{12}$$

Vehicles older than the maximum allowable age will be sold, while fleets can choose to replace those vehicles under the turnover age limit to minimize costs.

3.3. Vehicle Fueling and Charging Considerations

The method and equations introduced by Zheng et al. (2017) are implemented to determine feasibility of travel, e^s_{ij}, for vehicles originating at s over link (i, j), given the vehicle range and location of fueling and charging stations, as shown in Equations (13a)-(13f). As summarized by Zheng et al., the variable L^s_{qj} is incremented by the (i, j) link's distance d_{ij}, as shown in Equation 13a, if node i does not have fueling or charging infrastructure as identified by the parameter F_{qi}. The variable $e^s_{q,ij}$ will then have a value of 1 if the total distance L^s_i is within the vehicle's range limits. The location of stations is not optimized in our formulation; it is instead defined as a network parameter. Therefore the status, F_{qi}, of a node as a fueling or charging station is an input to the MILP.

$$L^s_{qj} \geq L^{s'}_{qi} + d_{ij} - M(1 - e_{q,ij}) \tag{13a}$$

$$L^s_{qi} \leq R_q \tag{13b}$$

$$L^{s'}_{qi} \geq L^s_{qi} - MF_{qi}, \quad L^{s'}_{qi} \leq L^s_{qi} + MF_{qi}, \quad L^{s'}_{qi} \leq M(1 - F_{qi}) \tag{13c}$$

$$x^s_{qn,ij} \leq Me_{q,ij} \quad \forall n,\ q \,\epsilon Q \tag{13d}$$

$$L^s_{qi} \geq 0, \ \ L^{s'}_{qi} \geq 0 \tag{13e}$$

$$e_{q,ij} \in \{0, 1\}, \quad F_{qi} \in \{0, 1\}, \quad F_{qi} = \begin{cases} 1 & i \in C \\ 0 & \end{cases} \tag{13f}$$

As stated earlier, fleet owners seek to increase the productivity of their vehicles in a cost-efficient manner to maximize revenue. Hours of service (HOS) rules limit both the driving time of commercial drivers to 11 hours and their time on-duty to 14 hours per day with a minimum 30 minute break every 8 hours (Federal Motor Carrier Safety Administration, 2013). As a result, any activity that causes downtime for the driver, and vehicle, within hours of operation will negatively impact productivity and, potentially, revenue. The inconvenience of lower vehicle range and longer fueling or charging stops when compared to the diesel baseline, along with time constraints to complete a day's driving, may limit adoption of alternative technologies. Equations (14a)-(14b) show the daily vehicle miles traveled (VMT) range constraints given fuel or battery energy content, E_q, number of fuel or charging stops, $\Phi_{qn,c}$, and energy transfer during on-road charging in the case of battery electric vehicles. It is assumed that a vehicle leaves its city of origin s with a full fuel tank or charged battery. Equation 15 limits total operational time, that is driving plus fueling or charging stops, to remain within HOS limits, η, while range-extending stops are limited to those nodes with station availability as specified by Equation 16.

$$\sum_{(i,j)\in A} x^s_{qn,ij} d_{ij} \xi_{q,ij} \le E_q(1 + \Phi_{qn,c}) \tag{14a}$$

$$\sum_{(i,j)\in A} x^s_{qn,ij} d_{ij} \xi_{q,ij} \le E_q(1 + \Phi_{qn,c}) + \Phi_{CR} \sum_{(i,j)\in G} E_{CR,ij} x^s_{qn,ij} \quad \forall q, n \tag{14b}$$

$$\sum_{(i,j)\in A} x^s_{qn,ij} t_{r,ij} + \Phi_{qn,c} t_{en,q} \le \eta \tag{15}$$

$$0 \le \Phi_{qn,c} \le \sum_i \sum_j x^s_{q,ji} F_{q,i} \tag{16}$$

Here, $t_{r,ij}$ is the time spent driving, $t_{en,q}$ is the time needed to charge or fuel, η is the total window of operation, and $\Phi_{qn,c}$ is the number of times a vehicle n of type q stops to fuel or charge.

The wireless power transfer system for on road charging includes long primary windings installed under the road and secondary pick-up windings installed below the chassis of the electric vehicle (Stamati and Bauer, 2013). The power transferred can recharge the battery while the vehicle is moving; the transfer of energy is proportional to the power of the system and the time the vehicle is above the primary winding. Energy transfer, $E_{CR,ij}$, for a given charging road depends on the speed of the vehicle, u_{ij}, and the length, L_{ij}, of the route conditioned with windings, such that

$$E_{CR,ij} = P_{CR,ij} L_{CR,ij} \frac{1}{u_{ij}} \ . \tag{17}$$

3.4. Traffic Model and Emissions

A macroscopic traffic model (Federal Highway Administration, 2017) is implemented and solved annually in the simulation, as done in (Guerrero et al., 2018), to compute the traffic flow speed based on the total number of passenger and freight vehicles, $q_{f,ij}$, introduced to the link such that

$$-\frac{k_{ij}}{v_{f,ij}} u_{ij}^2 + k_{ij} u_{ij} - \frac{q_{f,ij}}{N_{ij}} = 0. \tag{18}$$

Vehicle efficiency, $\xi_{q,ij}$ is then computed as a function of average vehicle speed over a network route (Transportation Research Board and National Cooperative Freight Research Program, 2010). Route time is computed as a function of traffic flow speed as $t_{r,ij} = \frac{d_{ij}}{u_{ij}}$.

In order to adequately project global emissions given adoption and utilization of vehicle technologies, the analysis must determine the emissions resulting from transportation, production, distribution, and burning of the fuels. A well-to-wheel analysis provides an estimate of the emissions involved from extraction of the source of energy (well) throughout its point of utilization (wheels) (Ramachandran and Stimming, 2015). Here, we distinguish between the well-to-tank, $\Psi_{wtt,q}$, emissions resulting from production, transportation, and distribution of fuel, and the tank-to-wheel values, $\Psi_{ttw,q}$, emissions resulting from the burning of the fuel. Both parameters have units of $\frac{kgCO_2}{EN}$, where *EN* represents the gallons of fuel or kWh of energy consumed. The annual regional emissions produced by vehicles of type q, $CO_{2,q}$, is computed as

$$CO_{2,q} = \sum_{(i,j)\in A} x_{q,ij} d_{ij} \xi_{q,ij} (\Psi_{wtt,q} + \Psi_{ttw,q}). \tag{19}$$

4. Powertrain Adoption Scenario

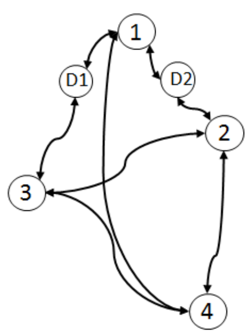
In this section we consider a single powertrain adoption scenario to demonstrate the predictive capabilities of the model under status quo assumptions; we will refer to this as powertrain adoption scenario A. Section 4.1 describes the calibration of the model to capture current public policies, short-term energy cost projections, freight demand growth, and limited availability of infrastructure for alternative fuels, over a period of 11 years, when evaluated policies are assumed to remain in effect. In Section 4.2 the model is used to project adoption and utilization strategies for emerging vehicle technologies as well as the potential to reduce CO_2 emissions under the assumptions presented in Section 4.1. Policies include daily hours of service (HOS) restrictions, GHG Phase 2 regulations, and economic incentives for the use of alternative fuels.

4.1. Scenario Definition

The model is calibrated to project adoption of six vehicle architectures with conventional and emerging powertrain technologies—diesel, CNG, LNG, hybrid electric diesel (HEVD), battery electric (BE), and hydrogen fuel cell (HFC)—by line-haul fleets operating over a small hypothetical regional network given a set of assumptions for the evolution of the FTS. The network is defined as a representative set of line-haul highway corridors, each under 500 miles long, connecting 4 city nodes and 2 distribution centers, as shown in Figure 4. In this scenario, it is assumed that availability of fueling stations for CNG, LNG, and hydrogen, as well as charging stations capable of servicing heavy-duty Class 8 BE vehicles, will be limited in the near future. LNG, hydrogen, and charging stations are only assumed for city 1, while CNG stations are located in cities 1 and 2 and diesel stations are modeled in all city nodes. Fueling and charging times are assumed as follows: 1) 0.2 hours for a diesel or hybrid-diesel vehicle, 2) 1.5 hours for a CNG vehicle, 3) 0.25 hours for an LNG vehicle, 4) 3 hours for a BE vehicle, and 5) 0.4 hours for a HFC vehicle.

The U.S. Department of Energy, through its Alternative Fuels Data Center (AFDC), and the U.S. Energy Information Administration (EIA) provide reports of historical as well as short and long-term projections for retail prices of diesel and alternative fuels (U.S. Department of Energy, 2018a; U.S. Energy Information Administration, 2018). Future fuel and energy cost trends indicated by the EIA are used to project costs of diesel and electricity from 2018-2028. For simplicity, historical prices for CNG and LNG, as published by the AFDC, are assumed for this projection scenario, given their retail price stability relative to diesel fuel costs. In the case of hydrogen fuel, stations servicing transit line heavy-duty buses in California report costs between \$8.00 and \$8.60 per kg of hydrogen dispensed for owning and maintaining stations as well as dispensing hydrogen to buses (California Fuel Cell Partnership, 2016). Therefore, a constant cost of \$8.00 per kg is assumed for the representative region modeled throughout the projected period of time for this adoption scenario. All fuel and energy costs over the projected time period are shown in Figure 5.

The following current policies are assumed to be in effect throughout the period of study: 1) Hours-of-Service (HOS) regulations limiting the driving time and operational time per day for property-carrying drivers (Federal Motor Carrier Safety Administration, 2013), 2) GHG Phase 2 fuel efficiency standards for heavy-duty trucks (U.S. Environmental Protection Agency, 2016), 3) 12% refundable fuel tax credits on compressed natural gas consumed based on those offered in the state of Indiana, and 4) \$30,000 tax credit on CNG and LNG fueling equipment based on that currently offered in the state of Indiana (U.S. Department of Energy, 2018b). GHG Phase 2 regulations are assumed to have an effect on fuel efficiency and cost of diesel and natural gas vehicles; therefore the effects of the policy are modeled accordingly as indicated in Table 7 and Figure 6. Policies providing incentives for alternative fuel vehicles in the form of tax credits, such as those modeled here for CNG and LNG fuels, are not available in every state in the U.S. Fleets operating over a cross-state regional network will only get reimbursed for the fuel purchased where the subsidy exists. In order to model this effect, natural gas credits are assumed to be limited to City 2 in this adoption scenario.



Route Distance Between Node Pairs (mi)

	City 1	City 2	City 3	City 4	D1	D2
City 1	-	182	297	470	148	91
City 2	182	-	242	289	-	91
City 3	297	242	-	309	148	-
City 4	470	289	309	-	-	-
D1	148	-	148	-	-	-
D2	91	91	-	-	-	-

Figure 4 & Table 6: Representative line-haul network with 4 city nodes and 2 distribution centers. All direct routes between nodes are less than 500 miles.

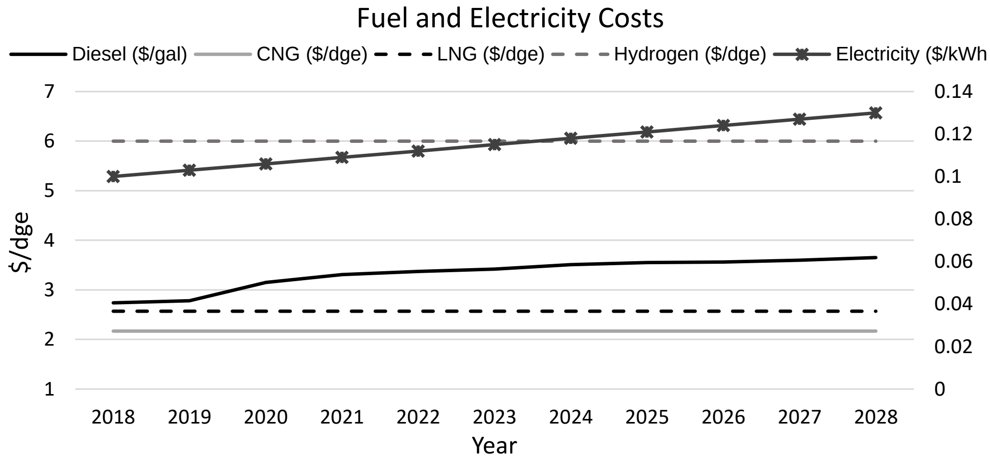


Figure 5: Assumed fuel (in diesel gallon equivalent) and electricity (shown on the right axis) costs for the period of time evaluated.

Table 7: Summary of GHG-2 Heavy-Duty Expected Fuel Savings and Price for Vehicle Model Year

	MY 2021	MY 2024	MY 2027
Maximum Tractor Fuel Savings (%)	13	20	25
Increase in Typical Tractor Price (%)	6	10	12

Values obtained from (U.S. Environmental Protection Agency, 2016)

The authors have previously identified a representative set of 12 line-haul fleets operating over a 4-city network and varying in size (number of vehicles), budget, annual growth, TCO outlook, and vehicle turnover periods (Guerrero et al., 2018). The selected fleet management parameter values, shown in Table 8, were identified by emulating historic adoption trends for widely available Class 8 diesel vehicles with enhanced aerodynamic technologies. Results showed that the parameterized model predictions matched historical adoption trends within an error of 10%. The same set of representative fleets is therefore used here to capture heterogeneous fleet behavior as we project adoption of emerging Class 8 powertrain technologies given the set of assumptions for the evolution of the FTS described. Cargo demand, by weight, is parameterized to represent three fleet sizes as indicated in Table 9. The freight demand ratio between all 4 cities is equivalent to that between Chicago (1), Indianapolis (2), St. Louis (3), and Nashville (4), a representative network of high-traffic freight corridors in the U.S. Midwest (U.S. Department of Transportation, 2018).

Table 8: Multi-Fleet Parametrization (Guerrero et al., 2018)

Fleet	Gamma(γ) years	Cargo (b_i^h) ton	Turnover years	Budget (B_f) $	Cargo growth %/year
1	5	small	[2,4]	low	1%
2	5	small	[1,3]	low	1%
3	5	medium	[3,5]	high	0
4	5	large	[2,4]	low	2%
5	6	small	[3,5]	high	0
6	5	large	[1,3]	low	1%
7	6	small	[2,4]	high	1%
8	5	medium	[2,4]	low	1%
9	5	large	[3,5]	high	0
10	6	medium	[5,7]	high	0
11	5	medium	[1,3]	high	0
12	6	large	[5,7]	high	2%

Table 9: Cargo Demand (b_i^h) Parametrization

Small Fleet (ton/day)						
O/D	City 1	City 2	City 3	City 4	D1	D2
City 1	0	90	60	20	20	20
City 2	100	0	15	10	0	20
City 3	55	10	0	5	20	0
City 4	15	10	5	0	0	0
D1	20	0	20	0	0	0
D2	20	20	0	0	0	
Medium Fleet (ton/day)						
O/D	City 1	City 2	City 3	City 4	D1	D2
City 1	0	150	100	30	20	20
City 2	160	0	25	15	0	20
City 3	90	20	0	10	20	0
City 4	0	15	10	0	0	0
D1	20	0	20	0	0	0
D2	20	20	0	0	0	
Large Fleet (ton/day)						
O/D	City 1	City 2	City 3	City 4	D1	D2
City 1	0	235	160	50	50	50
City 2	260	0	40	25	0	50
City 3	140	30	0	20	50	0
City 4	0	25	10	0	0	0
D1	50	0	50	0	0	0
D2	50	50	0	0	0	

Fuel vehicle efficiencies for the alternative powertrain technologies are estimated in a diesel gallon equivalent based on fuel energy content, while BE vehicle efficiency is expressed in kWh/mi. Payload capacity is an important factor for freight vehicles and can be a metric upon which vehicle owners discriminate between technologies. CNG tanks are approximately 2-2.5 times heavier than diesel ones for the same energy capacity, while LNG vehicle tanks are almost 50% heavier (Reinhart, 2016). These weight effects are introduced into the model as a vehicle payload capacity parameter, shown in Table 10, where it is assumed natural gas vehicles have a decreased capacity as compared to diesel vehicles.

In the case of BE vehicles, Mareev et al. (2017) estimate that the engine, fuel tank, exhaust aftertreatment system, and diesel exhaust fluid (DEF) tank that become obsolete on a vehicle with a battery electric powertrain weigh approximately 3700 lb. This estimate, along with the added weight of a 600 kWh battery, assuming a 0.15-0.2 kWh/kg energy density, causes a decrease between 1 and 2 tons in payload capacity for a Class 8 BE vehicle with respect to the diesel baseline. BE vehicles are therefore assumed to have a payload capacity of approximately 2 tons lower than that of a diesel vehicle, as shown in Table 10. Finally, a pre-transmission parallel hybrid electric with diesel engine (HEVD) vehicle architecture is assumed as modeled by Zhao et al. (2013a) and an average increase in fuel efficiency is imposed over the diesel vehicle, also reflected in the vehicle parameterization table.

Figure 6 shows the efficiency for vehicles with diesel, BE, and HFC powertrain technologies. Vehicle efficiency $\xi_{q,ij}$ is defined as a function of vehicle technology and route speed. In the case of diesel and natural gas vehicles, it is assumed that efficiency will increase given the model year of the vehicle purchased, as indicated by the expected impact of GHG Phase 2 regulations. Moreover, CNG vehicles are assumed to have a loss of 15% efficiency relative to the diesel baseline, while a 12% loss is assumed for LNG vehicles (Bill Boyce, 2014; Reinhart, 2016). Therefore, the same trends in diesel vehicle efficiency with respect to speed and year of purchase are used for CNG and LNG, but with the assumed lower efficiency. BE and HFC vehicle efficiency curves are assumed to remain the same throughout the period of study, regardless of the year the vehicles are purchased.

Table 10: Vehicle Architecture Parameterization

Vehicle Type	Vehicle Cost ($)	Eff. (mi/EN) (@55 mph)	Range (mi)	Capacity (ton)	Maint. Cost ($/mi)	Reliability (% trips/year)	Emissions[g] (kg CO_2/EN) Well-to-tank	Tank-to-wheel
Diesel[a]	145,000	6	1000	25	0.15	1	9.45	10.16
CNG[b]	172,000	5.1	600	23	0.165	2	2.23	7.11
LNG[c]	190,000	5.25	1000	23	0.165	2	2.56	7.73
HEVD[d]	175,000	6.3	1100	24	0.158	5	9.45	10.16
BE[e]	210,000	0.39	300	23	0.175	5	0.63	0
HFC[f]	250,000	11	450	24	0.175	5	17.6	0

Vehicle cost, efficiency, range, and payload capacity values based on a range of sources for HD Class 8 tractors

[a] Approximate range assumed for a conventional diesel tractor with 200 gal fuel tank capacity (Reinhart, 2016; Zhao et al., 2013b; Fulton and Miller, 2015)

[b] Approximate range and payload capacity values based on a 140 dge fuel tank capacity (Reinhart, 2016; Fulton and Miller, 2015)

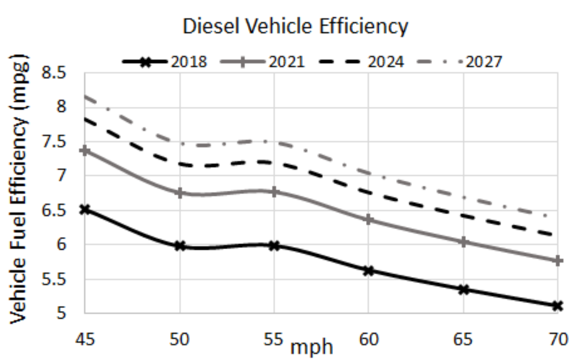
[c] Approximate range and payload capacity values assumed for a 270 dge fuel tank capacity (Reinhart, 2016; Zhao et al., 2013b; Fulton and Miller, 2015)

[d] Purchase cost, efficiency, range and payload capacity values assumed for a hybrid electric diesel tractor with 200 gal fuel capacity and 15 kWh battery capacity (Zhao et al., 2013b; Fulton and Miller, 2015)

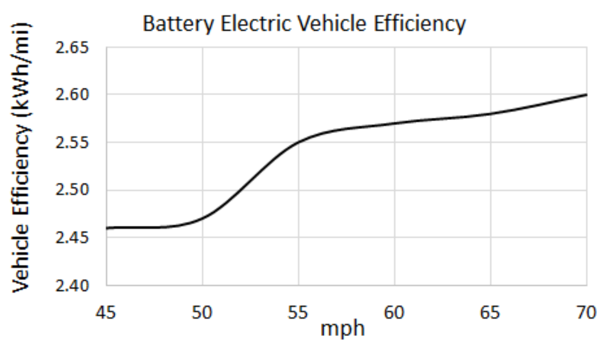
[e] Purchase cost, range, and payload capacity values assumed for a conventional tractor with a 600 kWh battery capacity. (Zhao et al., 2013b; Fulton and Miller, 2015; Mareev et al., 2017)

[f] Payload capacity and range assumed for a 65 kg fuel tank capacity and 11 mpg efficiency in diesel gallon equivalent (Zhao et al., 2013b; Fulton and Miller, 2015; Kast et al., 2017b,a)

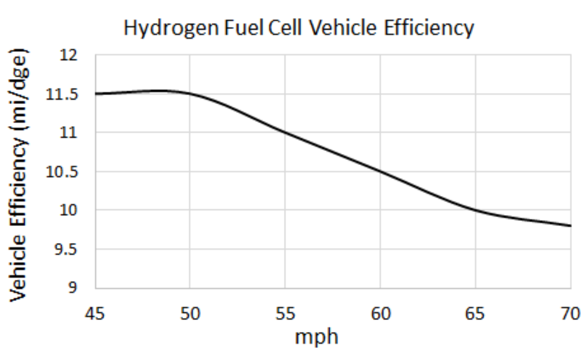
[g] Values based on US electricity mix, hydrogen produced by natural gas central reforming (Ramachandran and Stimming, 2015; Fulton and Miller, 2015)



(a) Diesel vehicle efficiency in mpg.



(b) BE vehicle efficiency in mi/kWh.



(c) HFC vehicle efficiency in miles per diesel gallon equivalent.

Figure 6: Vehicle efficiency as a function of vehicle speed.

4.2. Simulated Results

Following the assumptions and the model parameterization described in the previous subsection, we project displacement of diesel vehicles, adoption of alternative technologies, and CO_2 reduction over an 11 year period. This period of time was selected to identify the immediate effects of current and planned policies including GHG Phase 2 regulations, tax credits for alternative fuels, and hours of service (HOS) restrictions. The model is initialized with a 98% composition of diesel and 2% CNG vehicles across all fleets (Reinhart, 2016), with an assumption that all initially owned vehicles are either 1 or 2 years old. In order to quantify the potential in CO_2 reduction given the introduction and market penetration of alternative technologies, a baseline scenario is simulated assuming no changes in the current composition of the line-haul segment. That is, 98% of all vehicles used throughout the period of study are powered by diesel fuel, with only 2% of the vehicles operated using CNG. The CO_2 emissions produced in the baseline scenario are plotted as a dotted red line in Figure 7 for comparison.

Interesting effects can be observed with respect to adoption, traffic allocation, and most significantly, CO_2 emissions. Figure 7 shows the market penetration in the modeled region for all 6 vehicle architectures introduced. Given the assumed fuel costs, vehicle costs, limited infrastructure availability for alternative fuels, and continued effects of currently active tax credit incentives, diesel vehicles are primarily displaced, up to 80% in 2028, by CNG vehicles. There is limited adoption, less than 5% as shown in Figure 8, of LNG, BE, and HFC vehicles. There is no adoption of HEVD vehicles in this scenario. This analysis demonstrates a *rapid reduction in CO_2 in both the baseline and mixed adoption scenarios by the year 2021 due to the fuel efficiency impacts of GHG Phase 2 regulations, and an approximate reduction of 30% in CO_2 emissions (by mass) in 2028 once 80% of all vehicles used are powered by CNG fuel.* In this case, however, the immediate emissions reduction between 2018 to 2021 is higher, by comparison, in the mixed adoption scenario.

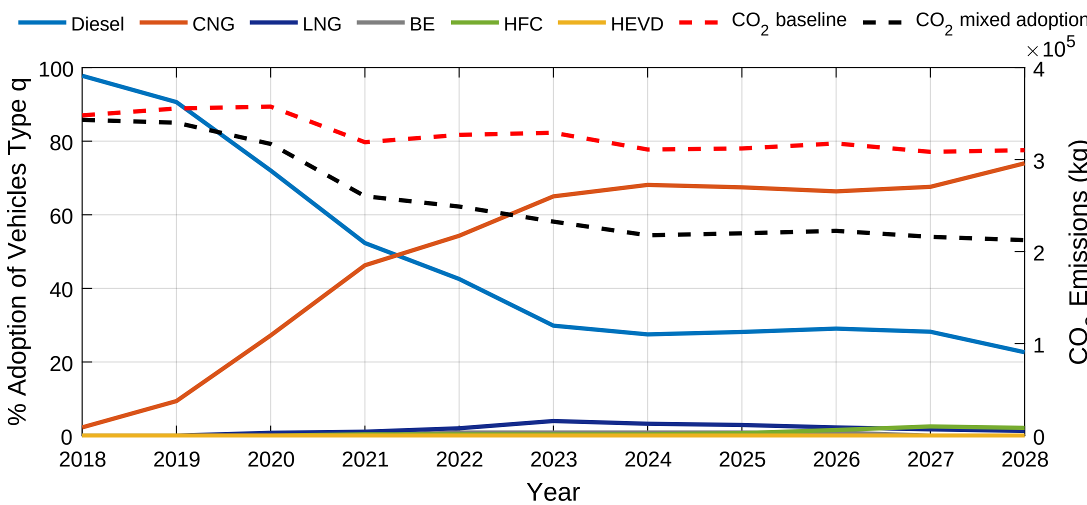


Figure 7: Total percent vehicle adoption throughout period of study across all regional fleets modeled. The red dotted line shows regional emissions for a baseline scenario with 98% diesel and 2% CNG fleet composition (see right y-axis). The black dotted line shows regional CO_2 emissions resulting from the projected adoption of the six vehicle architectures as shown on the same plot.

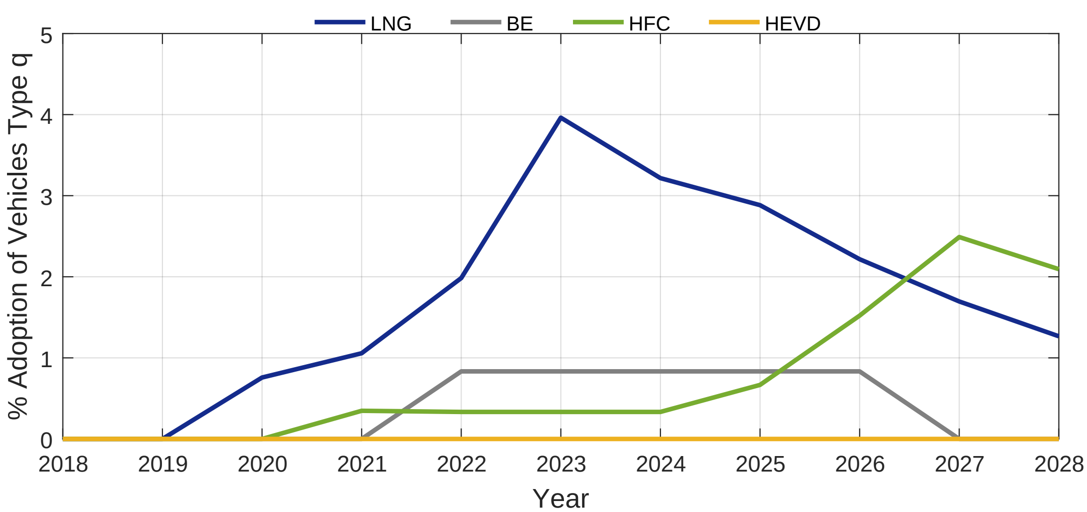


Figure 8: Adoption of LNG, BE, HFC, and HEVD vehicles throughout the scenario period.

As shown in Figure 7, the adoption rate for CNG reaches a plateau in the period 2023-2027, at which point we observe adoption of LNG and BE vehicles; this can be seen more closely in Figure 8. However, *GHG Phase 2 regulations are assumed to cause a further increase in fuel efficiency of diesel and CNG vehicles in 2027, causing an immediate increase* *in CNG adoption*. BE vehicles are no longer economically attractive to those few fleets that had adopted them and are therefore sold. Figures 9 and 10 show total daily vehicle utilization for all truck types and allocation over the freight routes. Figure 10 demonstrates a net increase in utilization of CNG vehicles along all routes over the period of study shown as adoption increases, with the exception of the longest route between city 1 and city 4 (Figure 10e), which sees a decrease in vehicle traffic throughout the period of study.

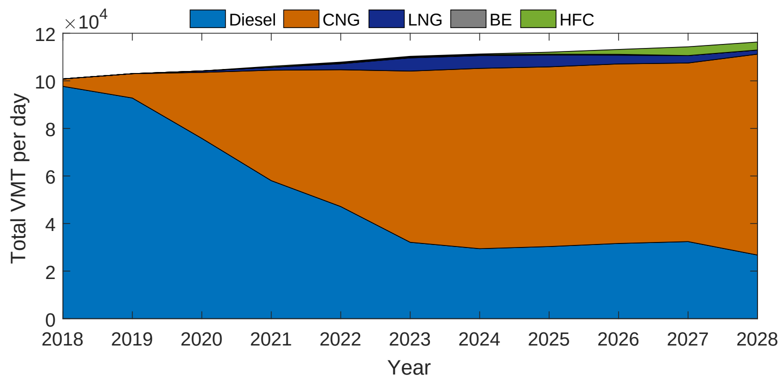


Figure 9: Daily total vehicle miles traveled (VMT) by all vehicle types.

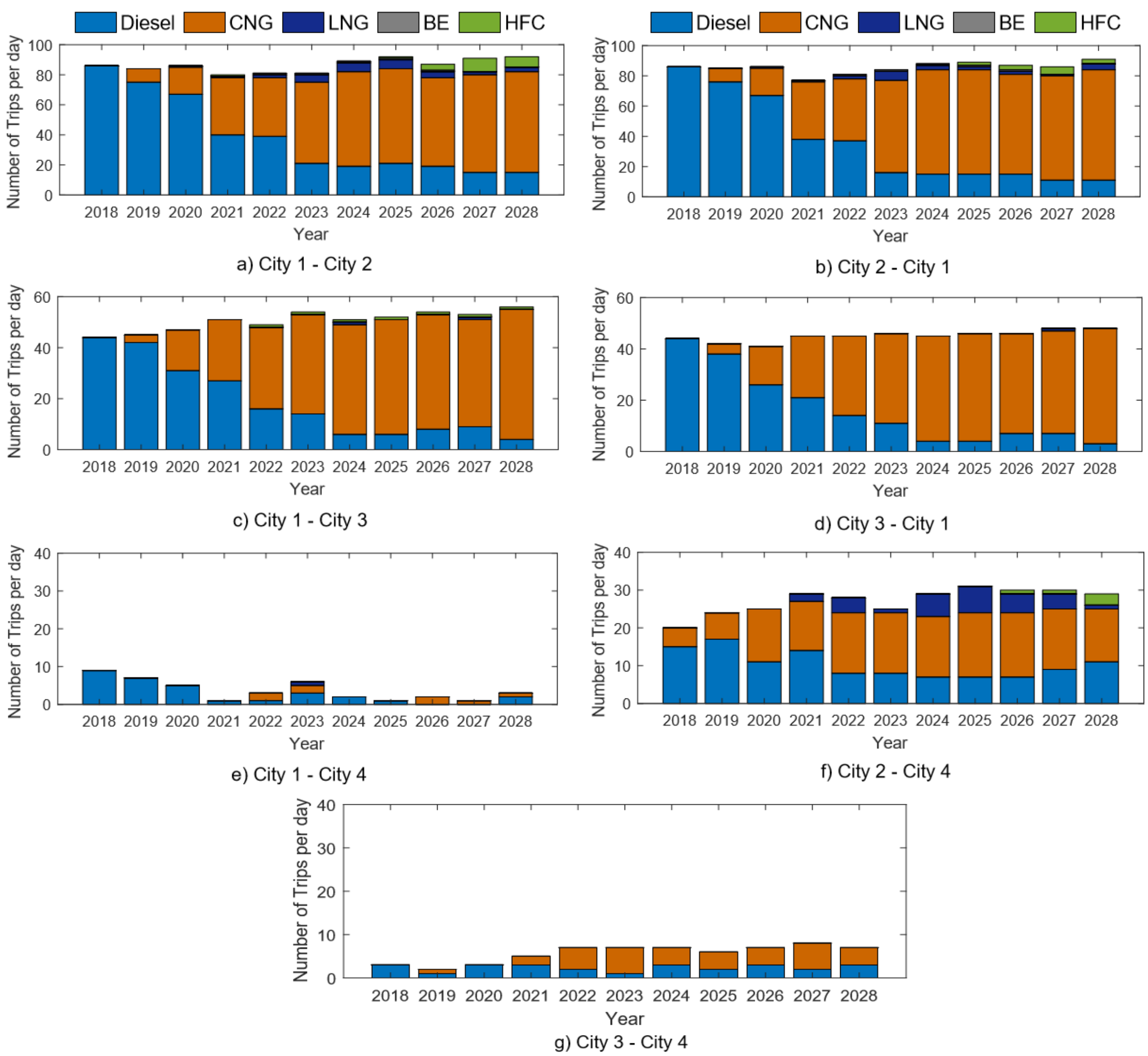


Figure 10: Total number of trips per day between selected cities, showing trips for all vehicle types adopted.

A higher adoption of CNG vehicles, which have a shorter range than diesel vehicles, could cause a shift in traffic *corridors as observed in this evaluation scenario.* This effect is shown in Figure 10. Here, the increase in adoption and utilization of CNG vehicles, which have a significantly shorter range as compared to diesel, causes a redistribution of vehicles transporting freight on the direct route from city 1 to 4 (Figure 10e) to reallocate through city 2 (Figure 10f) and city 3 (Figure 10g). At the beginning of the study period, 10 diesel vehicles travel from city 1 to 4; however, by 2021 and throughout 2028, fewer vehicles are taking this route. Alternatively, the routes between city 2 and city 4, and city 3 and city 4 see an equivalent increase in CNG vehicles after the year 2021. The route between city 2 to city 4 sees a small increase in LNG and HFC vehicle allocation after the year 2021.

5. Sensitivity Analysis

In this section, the proposed model is used to quantify the sensitivity of vehicle adoption and regional CO_2 emissions projections to variation in system factor assumptions such as future policies, fuel costs, proliferation of fueling/charging infrastructure, and technology design. Furthermore, we identify the mechanisms and adoption trajectories that result in the maximum reduction in regional CO_2 emissions.

5.1. Design of Experiments

The results presented in the previous section assumed a specific scenario, which we called powertrain adoption scenario A, for the evolution of the FTS, based upon a continuation of existing policies. However, a powerful use of the proposed model is for understanding *how* the adoption of emerging powertrain technologies, and the resulting line-haul system emissions, will be affected if the cost of fuel, policy incentives, or availability of fueling infrastructure varies from the status quo. Are there any vehicle design factors—efficiency, range, payload capacity—that can be improved to enhance adoption of cleaner technologies? Is there a required level of infrastructure proliferation necessary to drive adoption of alternative powertrain technologies? What incentives are necessary to improve the economic output of fleets as they adopt cleaner technologies, and how must they be introduced over a chosen timeline to influence adoption? Moreover, how is viability of adoption affected if several of these factors vary concurrently from the status quo scenario? To answer these questions, a design of experiments (DOE) is used to identify the factors that significantly influence multi-fleet adoption of the powertrain technologies and to quantify the variation of market penetration under different scenarios. Fifteen system parameters are varied as shown in Tables 11-14. A 180 point design of experiments (DOE) is defined with the JMP Statistical Analysis software (SAS Institute Inc., 2016) in order to reduce the number of executions from a full factorial design to screen the response and produce a satisfactory regression model for sensitivity analysis.

Line-haul fleets are generally concerned with a vehicle's *productive capability*, i.e. the amount of freight it will be able to transport in a given day, particularly when considering a new technology over the well-known capabilities of a conventional diesel vehicle (North American Council for Freight Efficiency, 2016). The perceived uncertainty in fuel economy, range, and payload capacity, particularly for alternative heavy-duty technologies—CNG, LNG, HFC, and BE—causes what is known as "range anxiety" (Zhao et al., 2013b). Variation in these vehicle-level parameters is introduced, as shown in Table 11. As heavy-duty Class 8 BE and HFC vehicles are yet to be introduced to the market, there is also perceived uncertainty on the future availability of fueling and charging stations, and their capabilities, to service line-haul trucks in a timely manner. Therefore, parameters affecting infrastructure are also considered as shown in Table 12. As of 2016, the only examples of hydrogen stations servicing heavy-duty vehicles are those in California for transit city buses, charging a vehicle with a capacity of 50 kg in approximately 10 minutes (California Fuel Cell Partnership, 2016). Given the expected tank capacity for a Class 8 truck to be above 60 kg for a range of 450 miles (Kast et al., 2017a), higher fueling times are introduced as shown in Table 12. In the case of BE vehicles, exploratory fast-charging values of half an hour are introduced. However, given that based on current technology projections, fast charging stations with the capability to charge a 600-900 kWh battery pack will likely take more than 1 hour, values greater than 1 hour are introduced as well.

Table 11: DOE Vehicle Parameter Levels

Truck Type	Efficiency[a] (mi/dge)		Range (mi)		Payload Capacity (ton)		
	Level 1	Level 2	Level 1	Level 2	Level 1	Level 2	Level 3
Diesel	5.35	5.85	1000	1000	25	25	25
CNG	4.5	5	600	600	23	24	23
LNG	4.65	5.2	1000	1000	23	24	23
HEVD	5.6	6.2	1100	1060	24	24	24
BE[b]	2.6	2.2	500	300	23	24	25
HFC	7.2	10	750	450	24	24	24

[a] Efficiency shown at a vehicle speed of 65 mph
[b] BE efficiency is defined in units of $\frac{kWh}{mi}$

Table 12: DOE Infrastructure Parameter Levels

Truck Type	Charging Time (hr)					Infrastructure Availability	
	Level 1	Level 2	Level 3	Level 4	Level 5	Level 1	Level 2
Diesel	0.2	0.2	0.2	0.2	0.2	all cities	all cities
CNG	1.5	1.5	1.5	1.5	1.5	city 1	all cities
LNG	0.25	0.25	0.25	0.25	0.25	city 1	all cities
HEVD	0.2	0.2	0.2	0.2	0.2	all cities	all cities
BE	0.5	1	2	3	3	city 1	all cities/roads
HFC	0.25	0.25	0.25	0.4	0.5	city 1	all cities

As stated earlier, three range-extending infrastructure modes have been proposed in support of BE vehicles: 1) charging stations, 2) battery swap stations, and 3) on-road charging. Therefore, the infrastructure type available in the region is imposed as a factor of variation in order to understand the effects on adoption of BE vehicles. Due to uncertainty in related costs to build, maintain, and service the infrastructure, variation is also introduced to the user fees modeled, particularly the fees and tolls related to the use of battery swap stations and on-road charging, as well as the increase in cost of purchase assumed for wireless power transfer capability. These values are shown in Table 13.

Table 13: DOE Battery Electric Vehicle Considerations

	Battery Capacity (kWh)	Battery Cost ($/kWh)	Infra. Mode Φ	Swap Station Fee ($/month)	Charg. Rd. Toll ($/trip)	Charg. Rd. Power (kW)	Charg. Rd. Length (%)	Charg. Rd. Veh. Upcharge ($)
Level 1	900	50	0	1500	10	50	0.2	1000
Level 2	600	100	1	3000	20	100	1	5000
Level 3		200						

Finally, as shown in Table 14, a natural gas fuel tax credit is applied to both CNG and LNG fuel. A new economic incentive is also introduced; a voucher, valid at the time of purchase, is made available for hybrid and zero emission vehicles (BE and HFC) similar to the ones provided by the California Air Resources Board Voucher Incentive Project (CARB HVIP) and currently available in the state of California (California Air Resources Board, 2018).

Table 14: DOE Regional Policies and Economic Factor Levels

Truck Type	Cost of Fuel ($/EN[a])						Policies	
	Level 1	Level 2	Level 3	Level 4	Level 5	Level 6	Level 1	Level 2
Diesel	Decrease[b]	2.8	Increase[c]	Increase	Increase	Increase	None	None
CNG	2.17	2.17	2.17	2.17	2.17	2.17	None	12% fuel tax rebate
LNG	2.57	2.57	2.57	2.57	2.57	2.57	None	12% fuel tax rebate
HEVD	Decrease	2.8	Increase	Increase	Increase	Increase	None	$20,000 voucher at purchase
BE	Increase[d]	Increase	Increase	0.11	Increase	Increase	None	$45,000 voucher at purchase
HFC	6	6	6	6	4	2	None	$45,000 voucher at purchase

[a] EN is expressed in units of fuel (gal or diesel gallon equivalent) or electricity (kWh) consumed
[b] Cost of diesel fuel is linearly decreasing between $2.7-$2.2 per gallon during period of study
[c] Cost of diesel fuel is linearly increasing between $2.7-$3.2 per gallon during period of study
[d] Cost of electricity is linearly increasing between $0.1-$0.14 per kWh during period of study

Figure 11 shows the total number of purchases, plotted by vehicle type, across all 180 experiments evaluated as part of the DOE study. Each experiment represents the total number of vehicle purchases across all fleets in the region throughout the 2018-2028 period, shown in descending order with respect to the total number of vehicles purchased. On average, approximately 500 new vehicles are purchased throughout the period of study for every DOE case evaluated. However, it can be observed that *the total number of vehicles purchased in the region increases with the number of battery electric vehicles adopted*, as these vehicles have a shorter range than the diesel baseline, while total freight demand in the region remains the same across all DOE cases. This effect is particularly true in the cases where battery swap stations are available and a service fee model is followed instead of direct ownership of BE vehicles, thereby reducing the total cost of ownership.

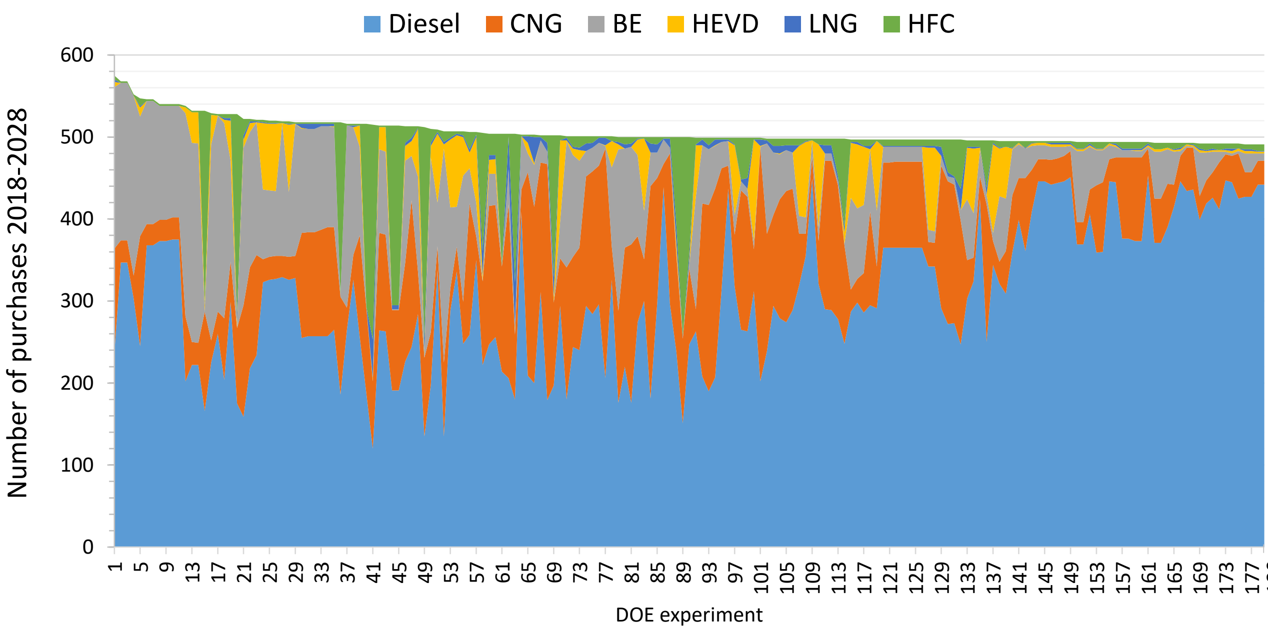


Figure 11: Number of purchases of each vehicle type throughout period of study across all DOE points evaluated.

Figure 11 shows that all cases evaluated result in mixed adoption scenarios of diesel, CNG, BE, HFC, and, to a lower extent, HEVD vehicles. There is negligible adoption of LNG vehicles given the factor values assumed for the DOE. Variation of individual adoption metrics to the factor levels evaluated are quantified in the appendix, showing a wide distribution of diesel, CNG, BE, HEVD, and HFC vehicle purchases. Furthermore, CO_2 emissions vary approximately between 2.62 Gg to 5 Gg across the different system evolution scenarios evaluated, as shown in Figure 12.

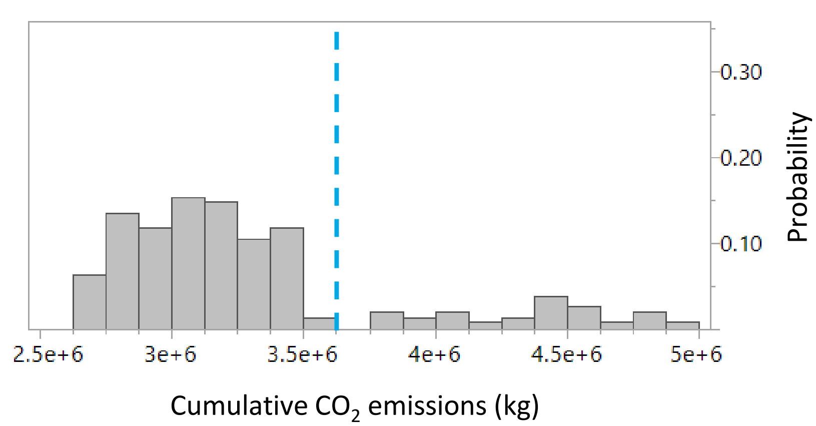


Figure 12: Histogram showing distribution of cumulative CO_2 emissions, in kg, produced throughout the 2018-2028 period as a result of the DOE factor variation. Cumulative emissions value for the baseline case (98% diesel adoption) is indicated by the blue dotted line.

In order to quantify sensitivity of vehicle adoption metrics to variation in individual factors, a regression model is computed from the DOE data and used to predict factor influence. The quality of fit for each regression model computed is shown in Table 15, indicating root mean square error and coefficient of determination (R^2) values. The R^2 values for diesel, CNG, HEVD, BE, and HFC purchase predictions, excluding LNG vehicles due to negligible adoption projections, are all above 0.87. Figure 22, shown in the Appendix, demonstrates the quality of fit of the regression models. The regression models are therefore a good representation of vehicle purchase projections, and here they are utilized to predict adoption sensitivity to factor variation. The effects are quantified in Figures 13–18.

Table 15: Summary of Regression Fit

Truck Type	R^2	RMSE
Diesel	0.87	31.6
CNG	0.9	22.
HEVD	0.97	6.86
BE	0.91	22.88
HFC	0.94	16.9

Tornado charts show individual effects of system factor variation, as indicated by the factor levels evaluated, in total number of vehicle purchases for each architecture. Recall that parameter values for individual factors are listed in Tables 11-14. Response sensitivity, shown on the y-axes of Figures 13-16, is computed as the deviation in vehicle purchases from the predicted mean as single factor levels are varied and all other factors are held constant.

In the case of diesel and CNG vehicles, as shown in Figure 13 and 14 respectively, *cost of diesel fuel is the most influential factor on projected adoption*. This is indicated by the positive effects of Levels 1-2 and negative effects of Levels 3-4 on diesel purchases. For this study, variation is introduced only for the cost of diesel, electricity, and hydrogen; exploratory values for the cost of natural gas were not considered given its stable price relative to the cost of diesel (U.S. Department of Energy, 2018a). Cost of electricity and hydrogen fuel are expected to vary regionally or with time as production efficiency progresses (Singh, M. et al., 2005; U.S. Energy Information Administration, 2018). Furthermore, although hydrogen as a transportation fuel is expected to remain high in cost in the near future, exploratory low cost values (Levels 5 and 6) are included in the study and also have a negative effect on diesel vehicle purchases. Moreover, an increase in payload capacity of alternative fuel vehicles—from 23 to 24 tons for CNG, LNG, and BE vehicles as indicated by Levels 1 and 2—has a negative effect on adoption of diesel vehicles while increasing the economic attractiveness of the alternatives (see Figures 15 and 16). In the case of CNG (Figure 14), the availability of battery swap stations (Level 2 of Φ_{BS} parameter) means that market penetration will be shared with battery electric vehicles (see Figure 16).

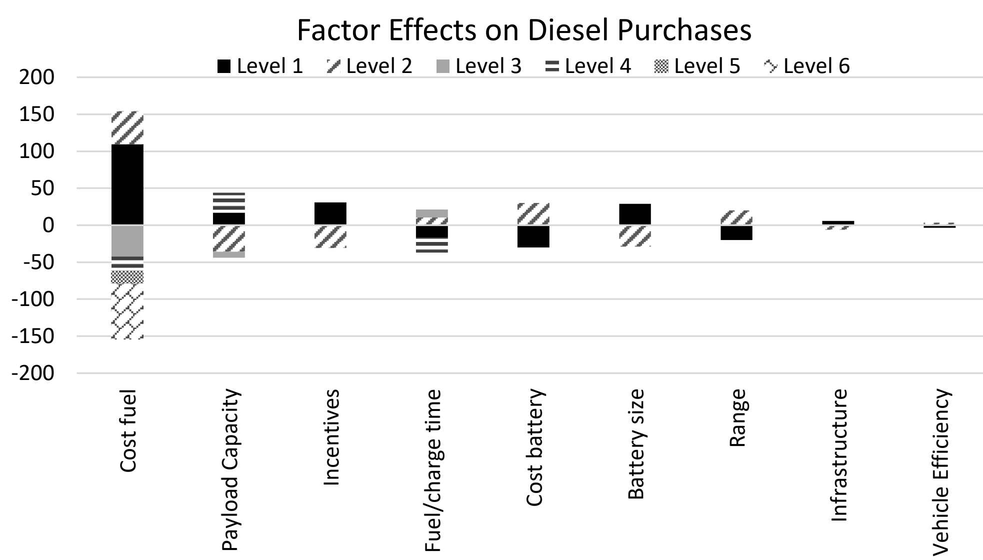


Figure 13: Tornado chart of factor effect on number of diesel vehicle purchases, shown as the deviation from the overall predicted mean of 317 diesel vehicle purchases.

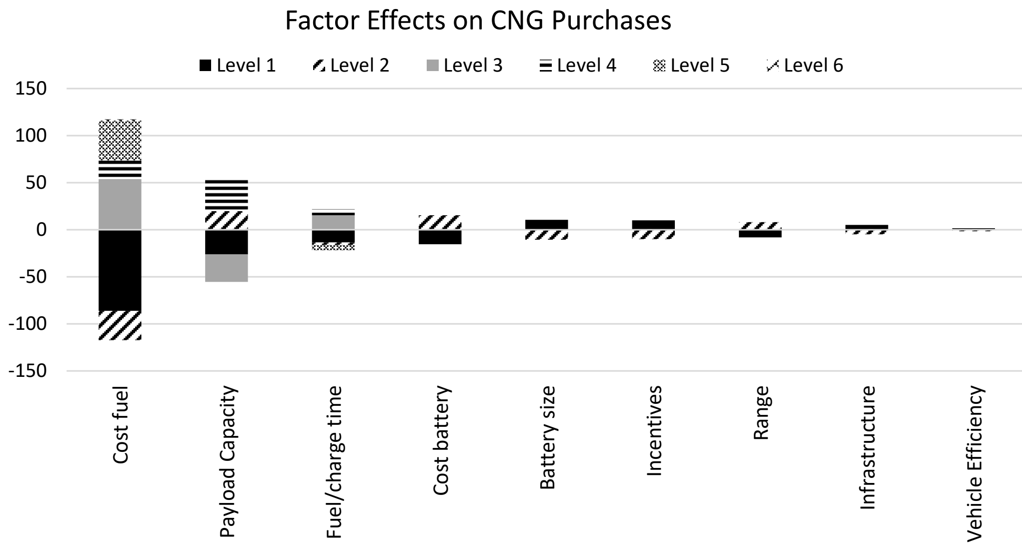


Figure 14: Tornado chart of factor effect on number of CNG vehicle purchases, shown as the deviation from the overall predicted mean of 86 CNG vehicle purchases.

Figures 15, 16, and 17 summarize factor effects on HEVD, BE, and HFC vehicle purchases, respectively. From Figure 15 we see that *incentives* are the primary factor influencing adoption of HEVDs, with a positive net effect if a voucher reducing cost of purchase by $20,000 is provided (see Level 2). Moreover, the availability of battery swap stations also causes a decrease in fleet preference of HEVDs, an effect also observed on adoption of CNG vehicles. Finally, Figure 16 corroborates the *positive effect of battery swap station availability on the attractiveness of BE vehicles* (see Level 2). As mentioned before, under this model, batteries are owned by service providers and a battery swap is assumed to take only 25 minutes. These assumptions significantly lower vehicle purchase costs, as fleet owners are in turn paying an annual flat service fee, as well as decrease the downtime associated with range-extending stops. Figure 16 also shows that variation in charging time, in the case where the use of charging stations is assumed, has a considerable effect on BE adoption. Specifically, faster charging times, 1 or 2 hours (Levels 1 and 2), have a positive effect on adoption as compared to the 3-hour baseline (Levels 3 and 4). *The imposition of on-road charging as the range-extending mode for electric vehicles appears to have a negative effect on BE vehicle adoption, as indicated in Figure 16, particularly when compared to the cases where battery swap stations are available.* This is due to the low power transfer capacity assumed possible, with a maximum value of 100 kW, not providing enough energy to fully charge the large battery packs necessary for Class 8 vehicles. In the case of HFC vehicles, Figure 17 shows the positive effect of a low cost of hydrogen fuel, demonstrating that values under $4 per diesel gallon equivalent are necessary for adoption of HFC vehicles.

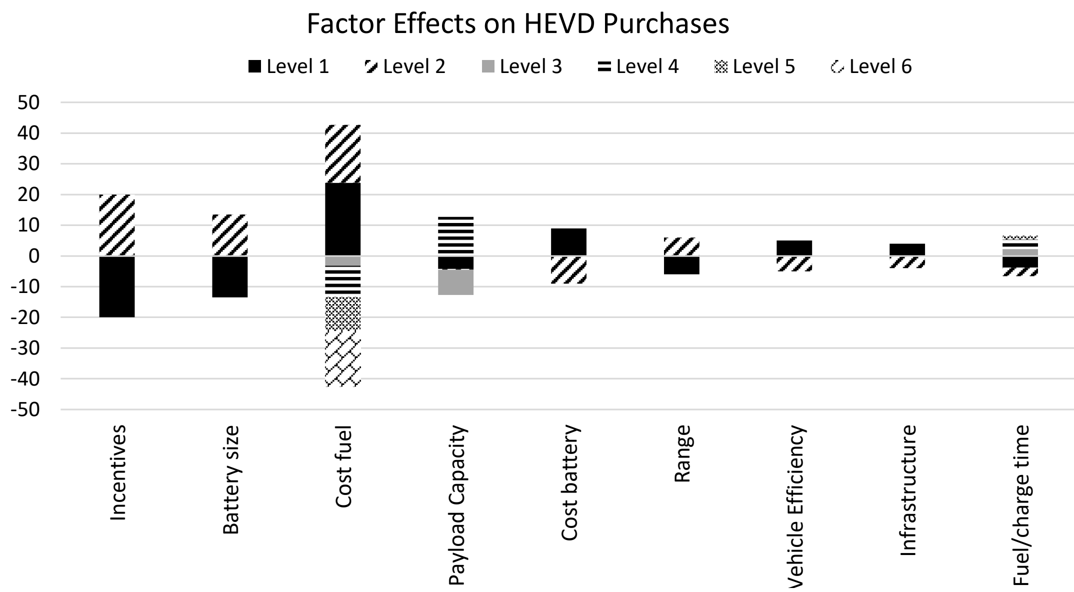


Figure 15: Tornado chart of factor effect on number of hybrid electric diesel vehicle purchases, shown as the deviation from the overall predicted mean of 19 HEVD vehicle purchases.

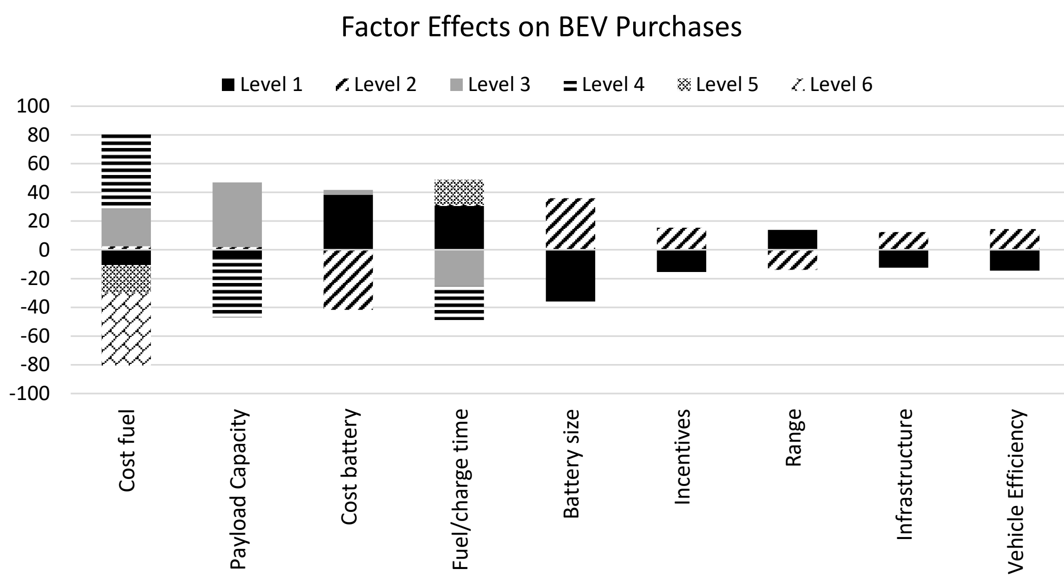


Figure 16: Tornado chart of factor effect on number of battery electric vehicle purchases, shown as the deviation from the overall predicted mean of 74 BE vehicle purchases.

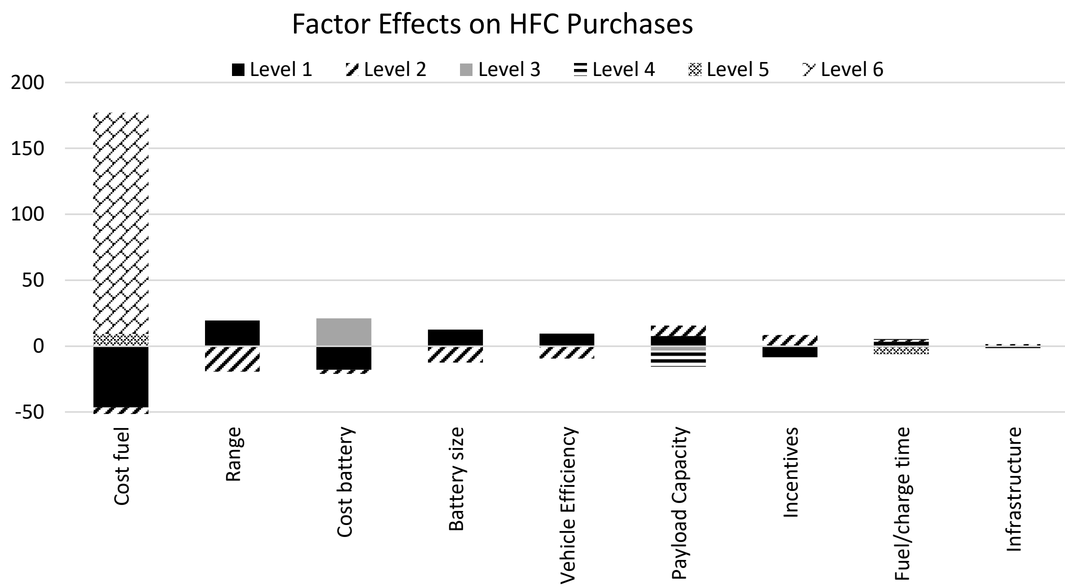


Figure 17: Tornado chart of factor effect on number of hydrogen fuel cell vehicle purchases, shown as the deviation from the overall predicted mean of 70 HFC purchases.

Due to the significant positive effect of battery swap station availability on BE adoption under these assumed scenarios, it is of interest to further inspect the effects of other parameters when this charging model is followed. We evaluate this case by setting $\Phi_{BS} = 1$ which imposes the use of battery swap stations according to the defined availability of infrastructure. In this case, as shown in Figure 18, *battery swap stations maximize the adoption of BE vehicles, provided that they are widely available*. As expected, adoption is also dependent on the service fee charged. Future work could focus on identifying an appropriate range in service fees, considering the costs of owning and operating a battery swap station for heavy-duty vehicles with large battery packs, to estimate the likelihood of such large positive effects on BE vehicle adoption.

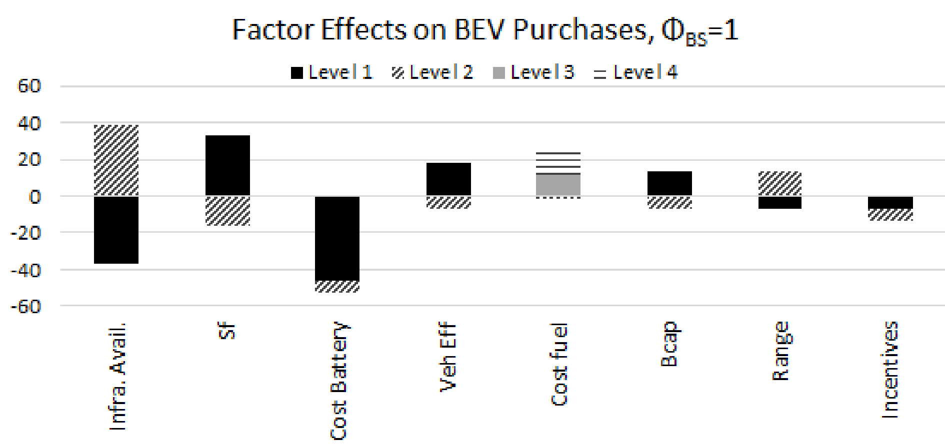


Figure 18: Tornado chart of factor effect on number of battery electric vehicle purchases when battery swap stations are used.

Finally, the direct effects of emerging vehicle penetration on regional CO_2 emissions are quantified in Figure 19. Here, perhaps most surprisingly, we observe that *an increased adoption of HEVD trucks has a negative impact on* CO_2 *emissions*, causing an increase of almost 2 Gg of CO_2 between the *zero adoption* and *maximum adoption* scenarios as indicated in Figure 19c. This happens particularly due to the higher vehicle utilization of HEVDs when adopted, an effect that is more pronounced when purchase vouchers are provided for this vehicle type. Here, it is important to note that only diesel-hybrid powertrains were considered. Future work could focus on exploring the effects on network wide emissions if other hybrid alternatives are introduced, for example, CNG-hybrid vehicles.

Moreover, cumulative CO_2 emissions in the region decrease as the number of CNG and BE trucks purchased throughout the period of study increases, as shown in Figures 19b and 19d. *As each of these vehicles reaches a volume of more than 200 vehicles adopted throughout the period of study, there is a potential reduction of approximately 20% in cumulative emissions from the baseline case (98% diesel adoption).*

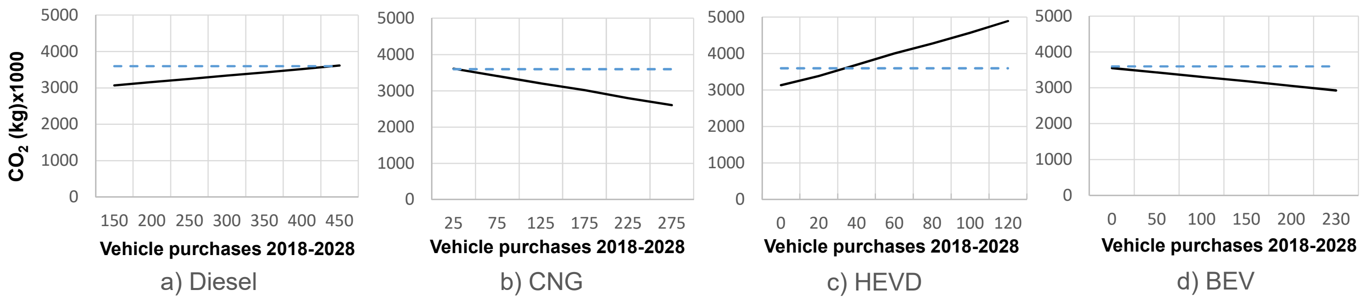


Figure 19: Cumulative CO_2 regional emissions, in kg, throughout the 2018-2028 period of study as a function of vehicle purchases. Cumulative emissions for the baseline case (98% diesel adoption) are indicated by the dotted line.

5.2. Paths towards lowering CO_2 emissions

Finally, we explore the mixed adoption trajectories, as identified by the DOE presented in Section 5.1, that minimize CO_2 emissions over the period of study in order to identify the mechanisms necessary to do so. Figure 20 shows the cumulative CO_2 emissions for all DOE points evaluated as a function of diesel vehicle purchases. As indicated previously by the sensitivity analysis, cases in which hybrid-diesel vehicles are heavily adopted result in high emission scenarios due to their over-utilization.

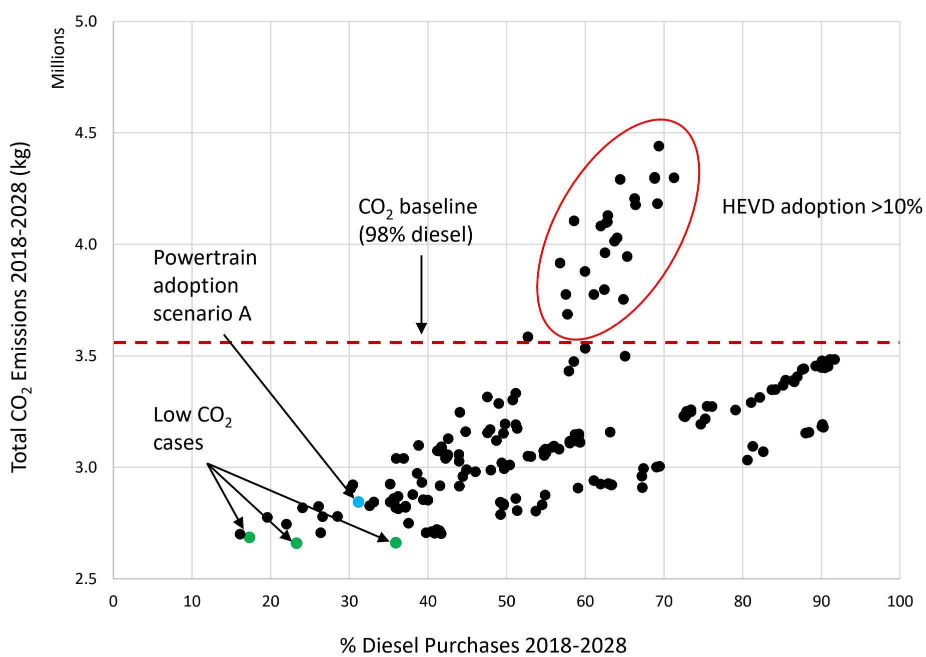


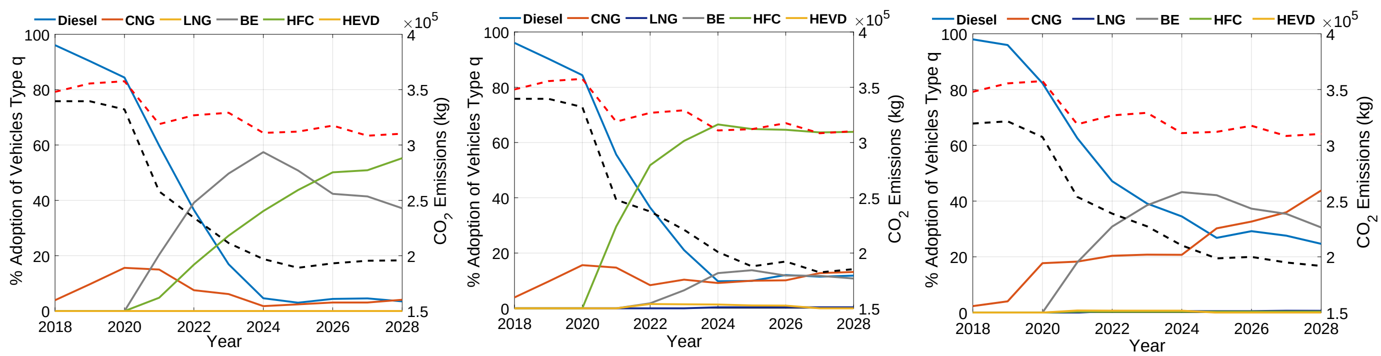
Figure 20: Cumulative CO_2 regional emissions shown for all DOE points, in kg, throughout the 2018-2028 period of study as a function of diesel vehicle purchases. Cumulative emissions for the baseline case (98% diesel adoption) are indicated by the dotted line.

Moreover, the three lowest CO_2 points are identified corresponding to significantly varying diesel and emerging technologies adoption scenarios, as shown in Figure 21 (a)-(c). Here, BE+HFC mix, primary HFC mix, and BE+CNG mix adoption cases are identified with similar low cumulative emission values. This indicates that *more than one future mixed-adoption scenario can be targeted in order to achieve a desirable reduction in emissions*. Table 16 shows the parameter levels corresponding to the three mixed-adoption cases; recall that factor levels are described in Tables 11-14. All three cases correspond to a wide availability of infrastructure, as indicated by Level 2 parameterization (Table 12).

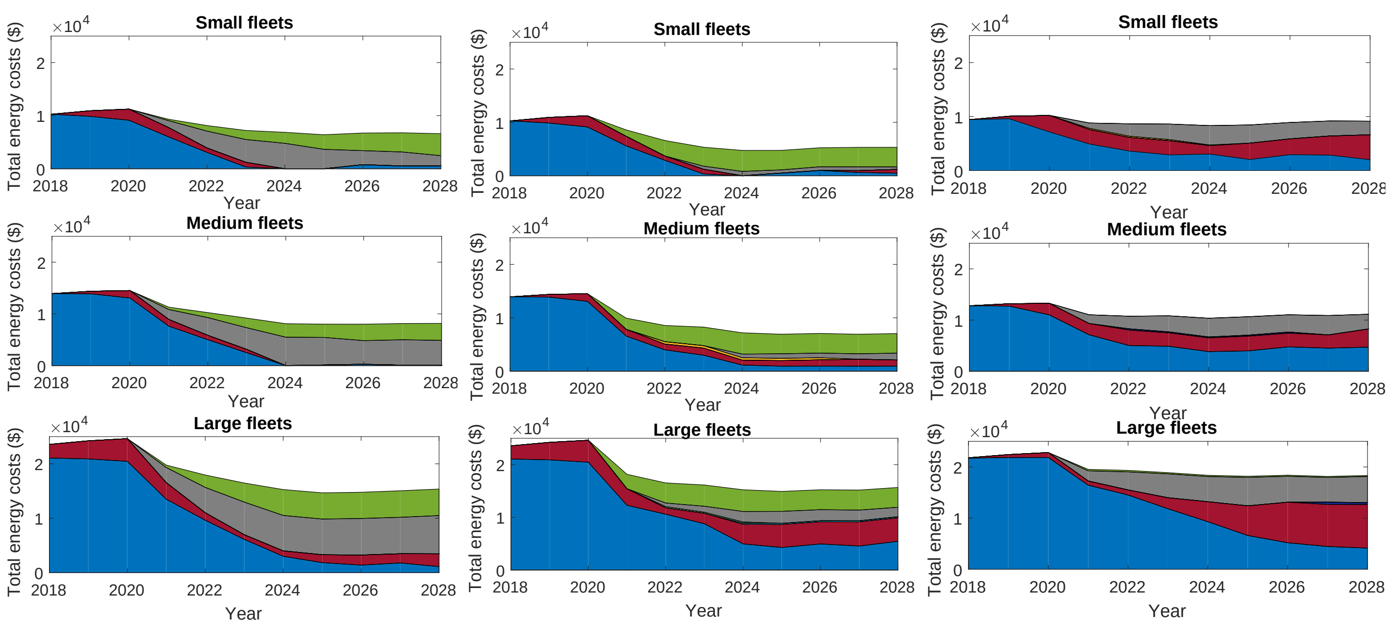
Moreover, the single difference in parameterization between cases 1 and 2 is the availability of incentives. However, case 2, in which incentives are not provided, shows that adoption of BE vehicles has instead been displaced by HFC and CNG adoption and a visible decrease in CO_2 emissions in 2028, as shown in Figure 21 (b). This indicates that case 2, as defined by the future parameter values, shows robustness to the removal of voucher incentives with respect to the capability to achieve a large reduction in cumulative emissions. Furthermore, cases 1 and 2 both show high adoption of HFC vehicles and correspond to the low cost assumption for hydrogen fuel of $2 per diesel gallon equivalent. This suggests that although voucher incentives may not be necessary, a fuel subsidy may be required in the future in order to achieve such a low cost for hydrogen fuel. It is cases 1 and 2, as shown in Figure 21 (d) and (e), that observe the highest reduction in total energy costs for all fleet types, as compared to the case 3 with mixed diesel, BE, and CNG adoption shown in Figure 21 (f).

Table 16: Low CO_2 Point Parameters

	Efficiency	Range	Payload Capacity	Cost of Fuel	Policies	Fueling/Charging Time	Infrastructure Availability	Cumulative CO_2 Emissions Reduction
Case 1	Level 1	Level 1	Level 3	Level 6	Level 1	Level 1	Level 2	25.3%
Case 2	Level 1	Level 1	Level 3	Level 6	Level 2	Level 1	Level 2	25.8%
Case 3	Level 2	Level 2	Level 2	Level 3	Level 2	Level 4	Level 2	25.9%



(a) Mixed adoption case 1 (b) Mixed adoption case 2 (c) Mixed adoption case 3



(d) Fuel cost case 1 (e) Fuel cost case 2 (f) Fuel cost case 3

Figure 21: Low CO_2 cases identified by the DOE study.

5.3. Summary

Here we summarize the key points and observations from the results presented in Sections 4 and 5. Powertrain adoption scenario A, presented in Section 4, demonstrated the potential for CNG vehicles to displace diesel by up to 80% in 2028 in a hypothetical line-haul network, assuming current public policies and tax credit incentives, short term energy and vehicle cost projections, and limitations of infrastructure availability for alternative fuels remain in effect. Furthermore, this analysis demonstrated that the continuation of GHG Phase 2 regulations could cause a rapid reduction in CO_2 in both the baseline (98% diesel adoption) and powertrain adoption scenario A by the year 2021. Finally, this scenario showed that a higher adoption of CNG vehicles, which have a shorter range than the diesel baseline, could cause a considerable shift in traffic corridors.

The DOE analysis presented in Section 5.1 showed that diesel vehicles are a majority of all trucks purchased throughout the 2018-2028 period in most evaluated DOE scenarios, with CNG and BE following in adoption volumes. A sensitivity analysis showed that adoption of HEVD vehicles, which was primarily incentivized by the provision of a $20,000 purchase voucher, caused a significant increase in CO_2 emissions in the region due to higher utilization of this vehicle type. Although these results would indicate that, under these circumstances, HEVD adoption is not an appropriate path to lower emissions, the introduction of other hybrid alternatives to the market, e.g. a hybrid-electric CNG powertrain, should be evaluated given the considerable reduction in emissions when CNG vehicles are widely adopted. Finally, widespread adoption of CNG, BE, or HFC vehicles, as indicated by the number of vehicles purchased throughout the period of study, showed a potential reduction in CO_2 emissions of approximately 20% from the baseline scenario (according to the regression model).

It was also shown that CNG vehicle adoption is highly dependent on the cost of diesel fuel. Future implementation of this simulation framework should focus on quantifying the uncertainty of CNG adoption projections due to the volatility of historical diesel prices. Adoption of BE vehicles was incentivized by a widespread availability of battery swap stations with a short swap time of 25 minutes, assuming a service fee model was followed. In the case where direct ownership and use of charging stations was assumed, a charging time of 2 hours or less was necessary for widespread adoption. Finally, adoption of HFC vehicles was shown to be enabled by a low cost of hydrogen fuel (under a $4 per diesel gallon equivalent).

A deeper analysis of the three lowest CO_2 cases identified by the DOE analysis was presented in Section 5.2 and suggested that more than one future mixed-adoption scenario can be targeted in order to achieve a desirable reduction in emissions. Three adoption cases: a BE+HFC mix, primary HFC mix, and BE+CNG mix were identified with similar low cumulative emission values, and an approximate reduction of 25% from the baseline case presented in Section 4.2.

6. Conclusion

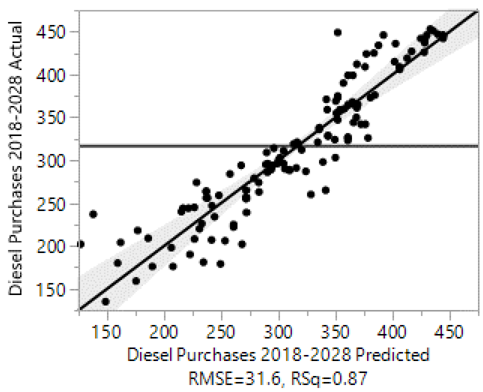
In this paper a constrained mixed-integer linear program was created to represent the decision-making process for heterogeneous fleets selecting vehicles and allocating them on freight delivery routes to minimize total cost of ownership. An SoS engineering methodology was used to define and abstract the line-haul freight transportation system (FTS), taking into account operational, economic, and policy factors that influence the economic attractiveness of conventional and emerging powertrain technologies and fleet adoption behaviors. The factors considered included vehicle efficiency and capacity, fleet budget and vehicle turnover periods, fuel and electricity costs, and availability of fueling and charging infrastructure in the network. This formulation was implemented to project alternative powertrain technology adoption and utilization trends for a set of line-haul fleets operating on a regional network. Alternative powertrain technologies include compressed and liquefied natural gas engines, diesel-electric hybrid, battery electric, and hydrogen fuel cell. After simulating the model with assumptions on model inputs that reflect the status quo, a DOE study was used to quantify sensitivity of market penetration projections to variation in vehicle performance parameters, infrastructure considerations, regional policies, and economic factors. Three mixed-adoption scenarios were identified with the potential to reduce cumulative CO_2 emissions by 25% for the period of study. Most importantly, the proposed model and framework can be used by several stakeholders, including fleet owners, powertrain manufacturers, and policy makers, to determine *how to shape adoption trajectories over a given time period* given variation in a broad range of factors that affect adoption.

Acknowledgment

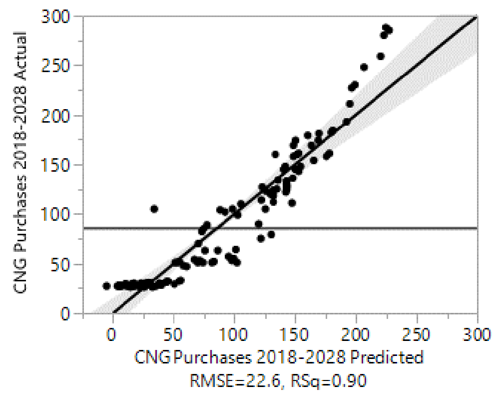
The authors thank Cummins Inc. for their support of this research.

Appendix

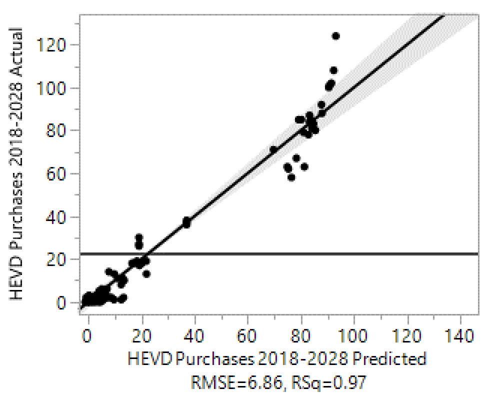
Figures 22 and 23 support the sensitivity analysis presented in Section 4.2. The quality of fit for the regression models computed in Section 4.2 are shown in Figure 22, indicating root mean square error and coefficient of determination (R^2) values for predicted vehicle purchases. Figure 23 shows the variation in vehicle purchases given the factor levels introduced in the sensitivity study.



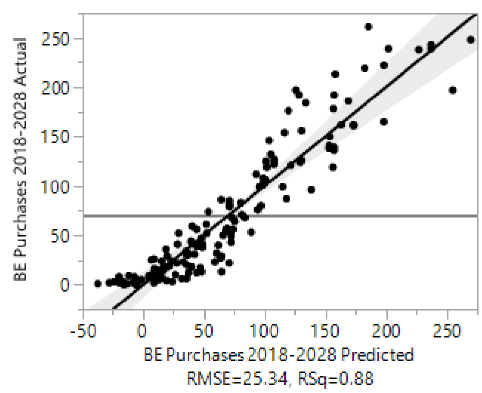
(a) Diesel Baseline



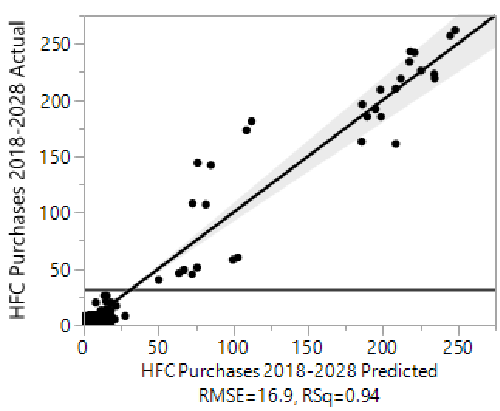
(b) CNG



(c) HEVD

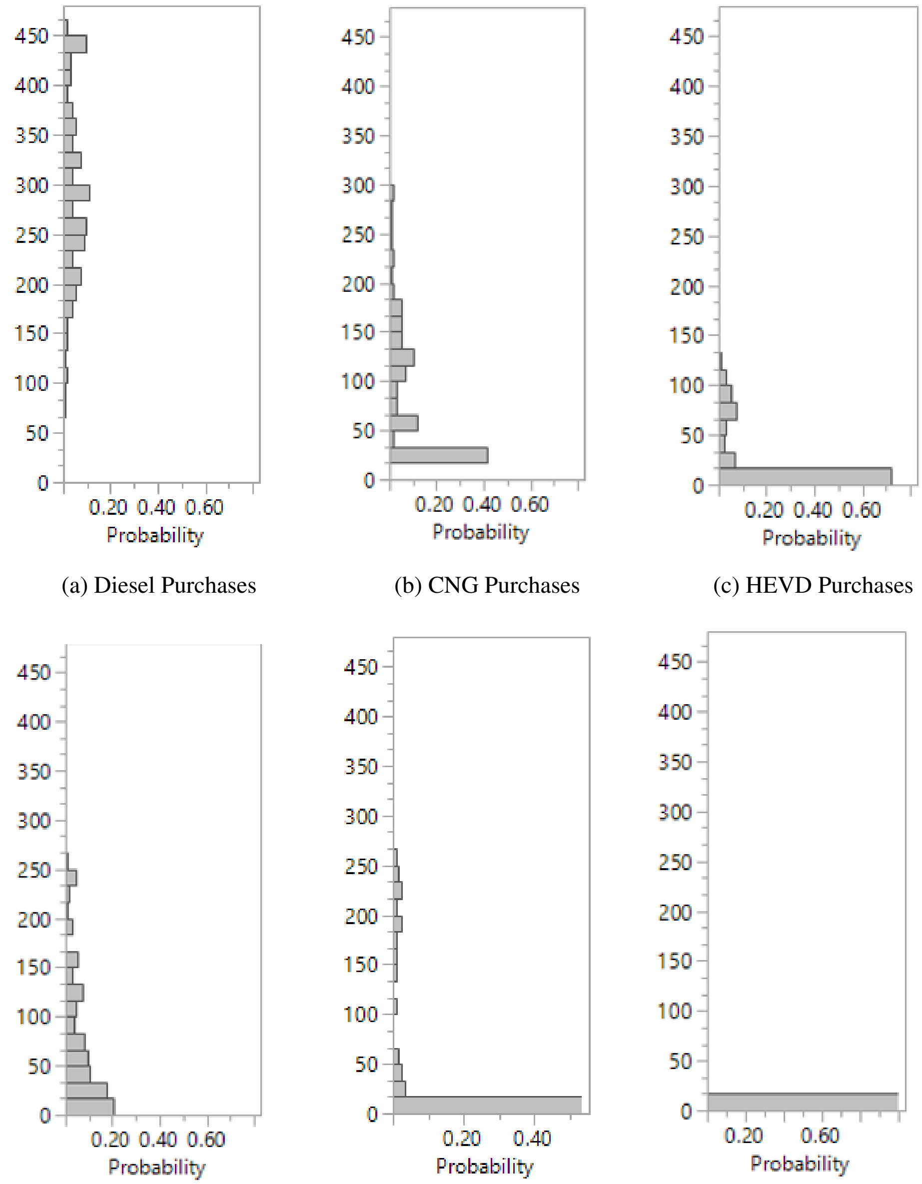


(d) BE



(e) HFC

Figure 22: Summary of regression fit showing mean of response, RMSE, and R^2 for predicted vehicle purchases throughout period of study.



(a) Diesel Purchases (b) CNG Purchases (c) HEVD Purchases

(d) BE Purchases (e) HFC Purchases (f) LNG Purchases

Figure 23: Histograms showing distribution of vehicles purchased throughout 11-year period as a result of the DOE factor variation.

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